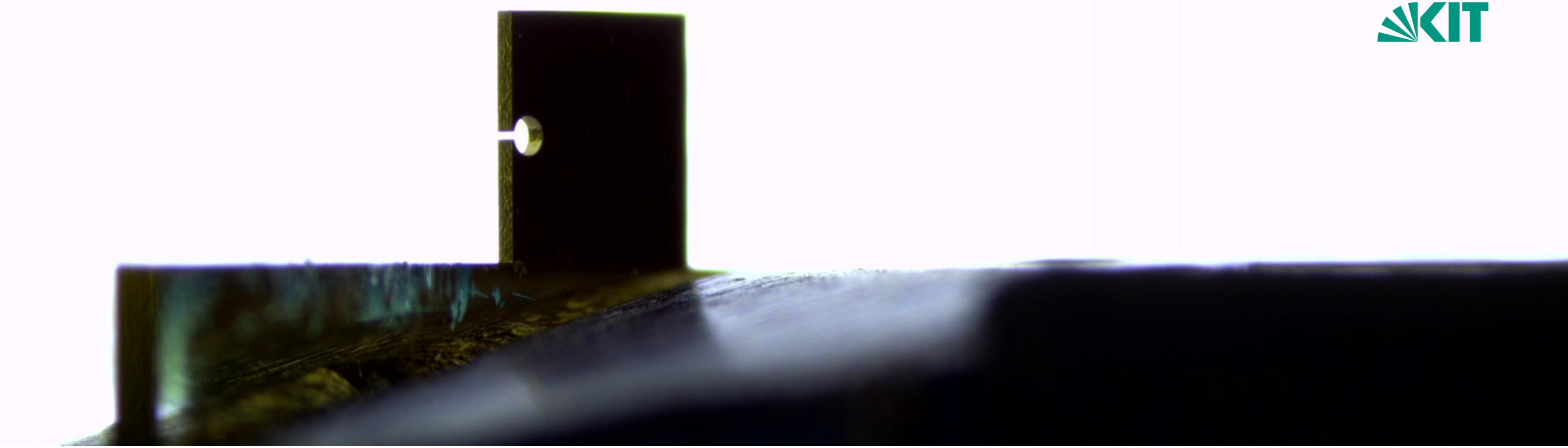


Terahertz Streaking Detection for Longitudinal Bunch Diagnostics at FLUTE

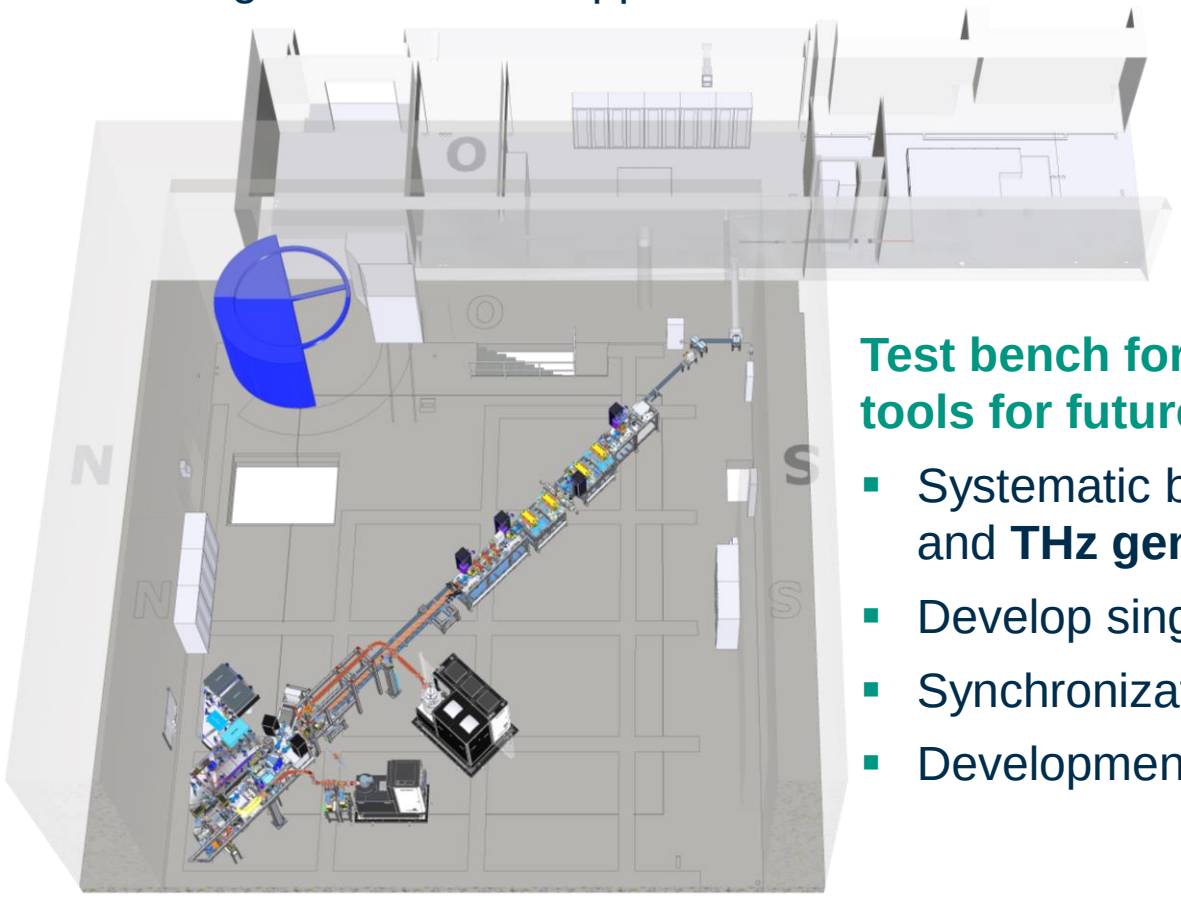
Matthias Nabinger
on behalf of the Compact-TDS collaboration

FLUTE: Accelerator Test Facility at KIT

Electron Energy	up to 90 MeV
Repetition Rate	up to 50 Hz
Electron Bunch Charge	up to 1 nC
Electron Bunch Length	down to 1 fs
Spectral Band Coverage	up to 30 THz
THz E-Field Strength	up to 1 GV/m

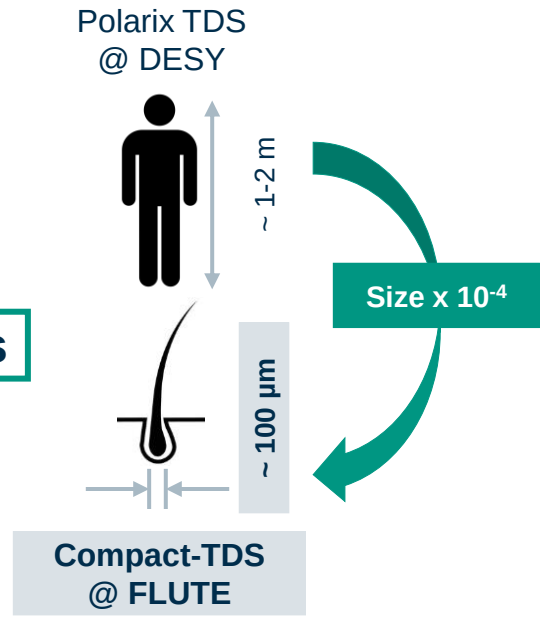
Short-pulse linear accelerator FLUTE

- **Test facility** for accelerator physics
- **THz** generation and applications



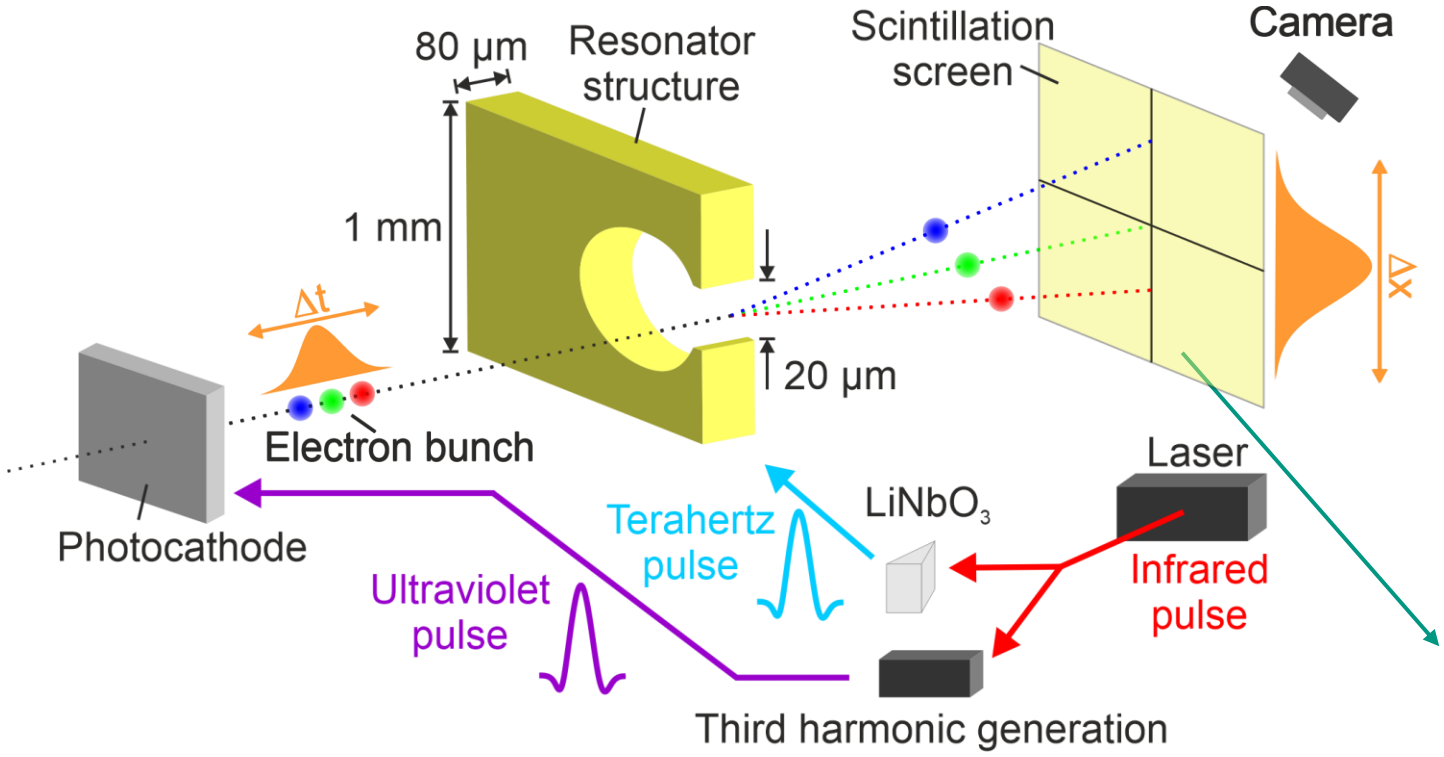
Test bench for new diagnostic methods and tools for future accelerators

- Systematic bunch compression and **THz generation studies**
- Develop single shot **femtosecond diagnostics**
- Synchronization on a **femtosecond** level
- Development of **AI Control**



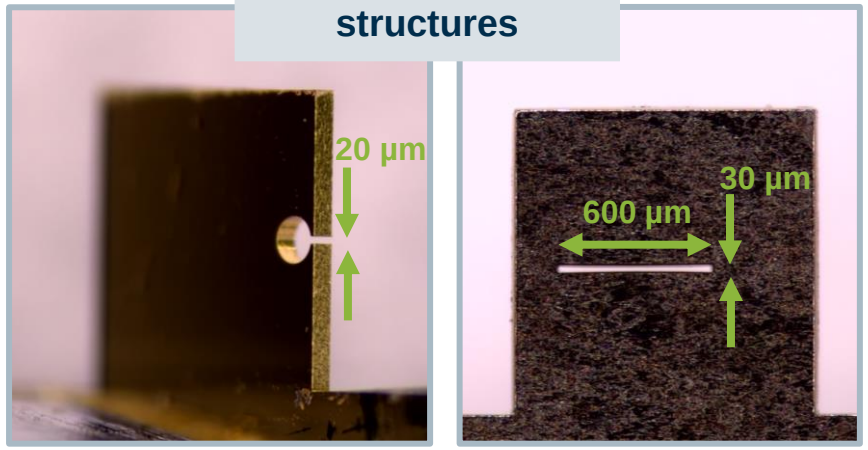
Compact femtosecond bunch profile diagnostics

THz-based transverse deflecting system

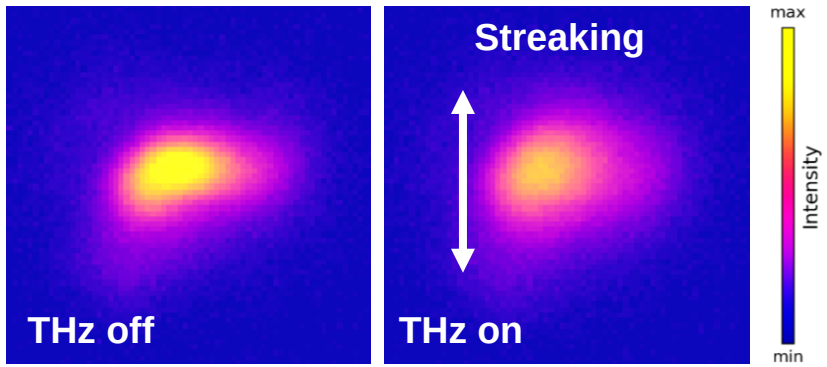


GHz → **THz** + **Resonator** amplification = **Compact & efficient** fs bunch profile measurement

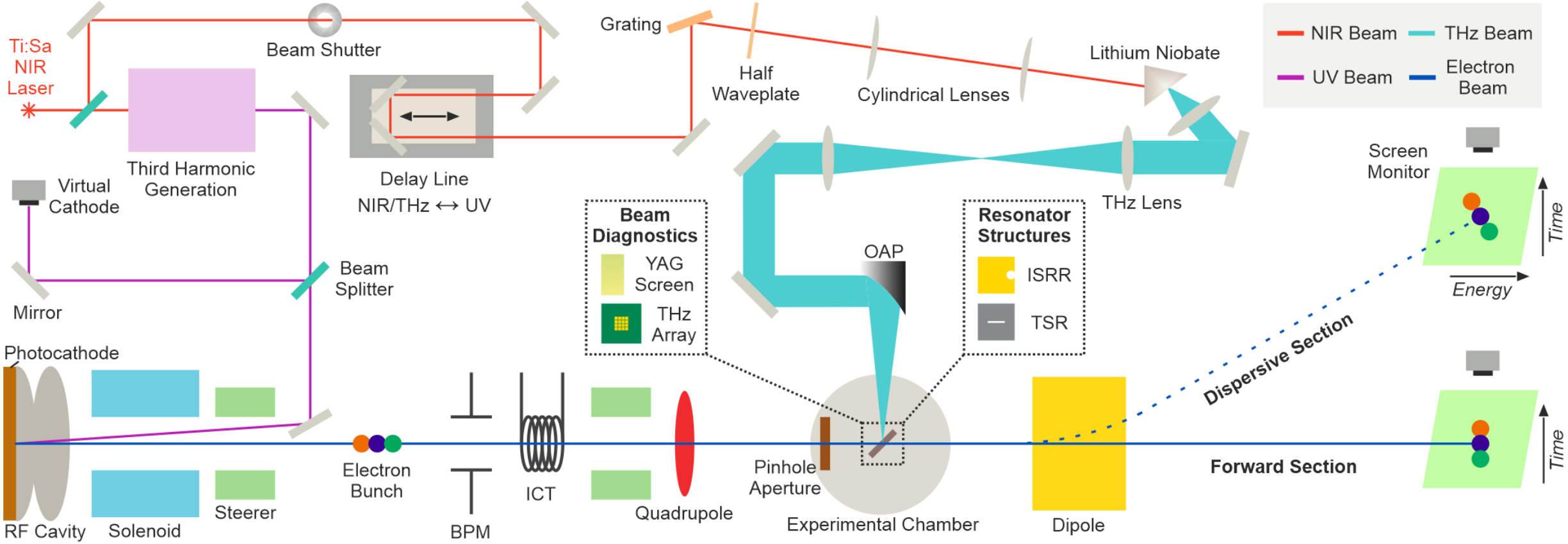
Resonator structures



Recently: streaked signal detected!

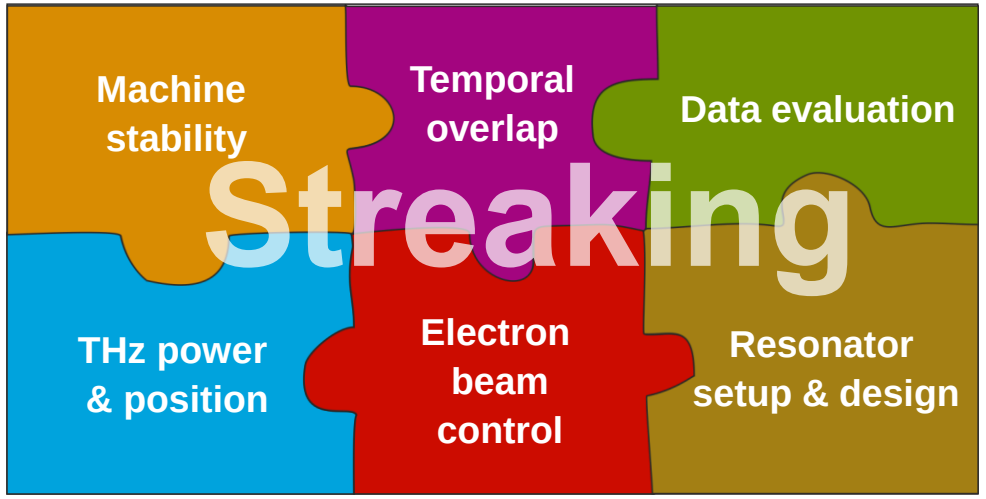


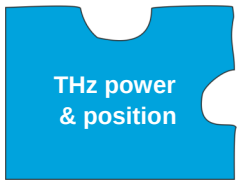
Compact-TDS setup – overview

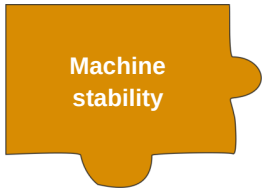


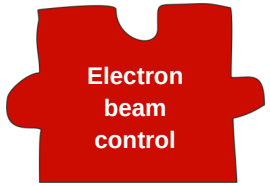
Solving the puzzle...

‚Matthias, which puzzle piece was it in the end that solved it?‘

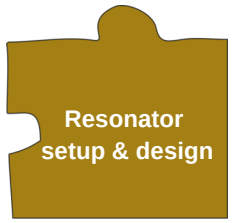






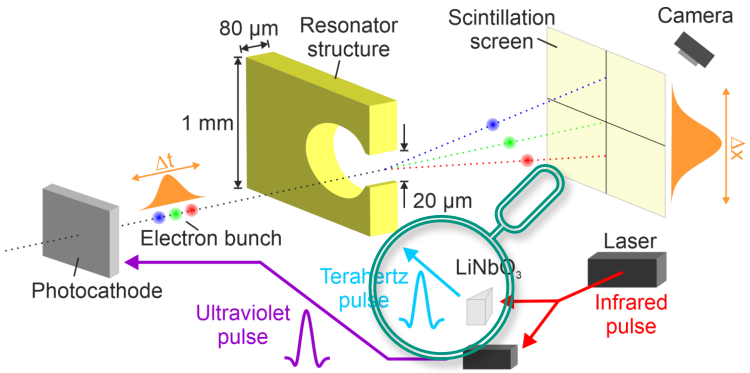


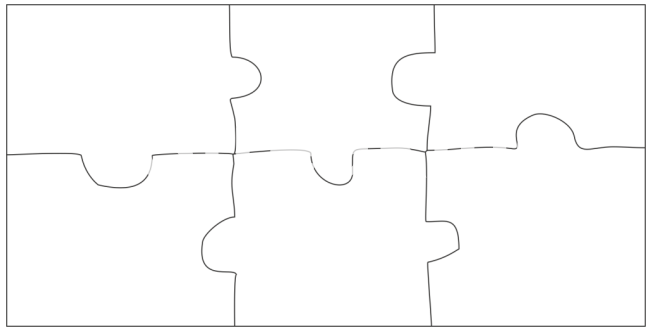




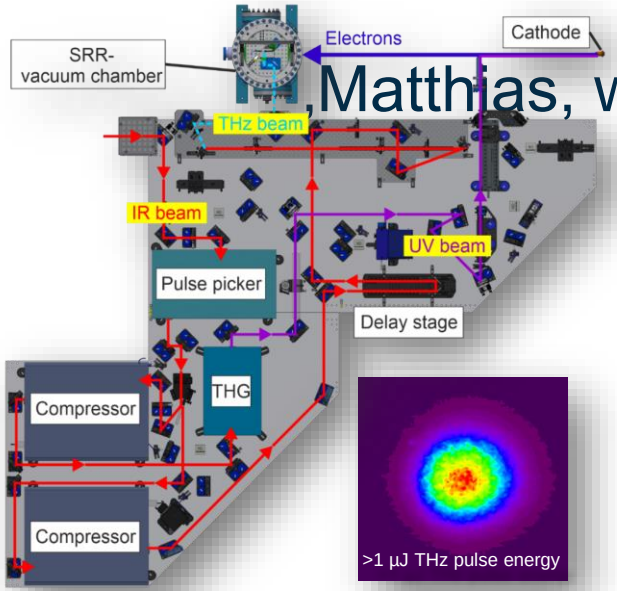


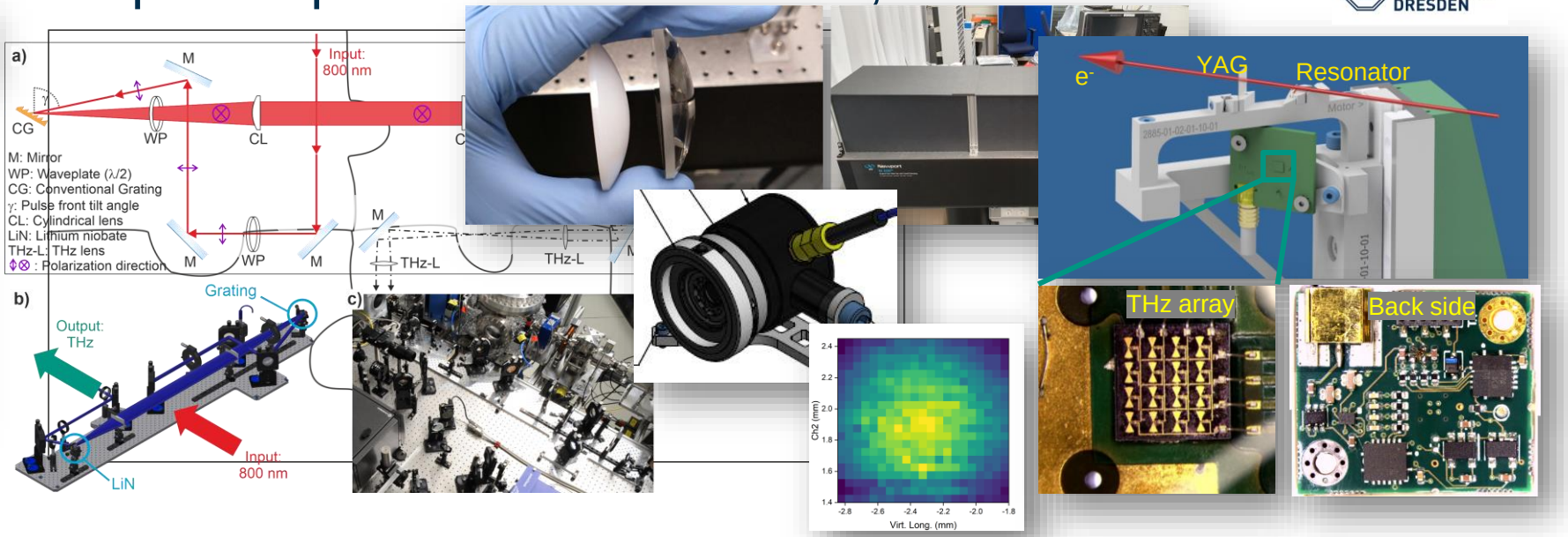
Puzzle piece: THz power & position





„Matthias, which puzzle piece was it in the end, that solved it?“





THz power & position

Machine stability

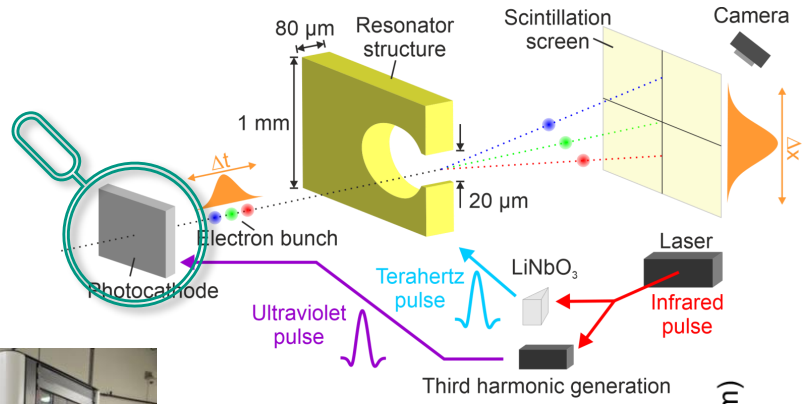
Electron beam control

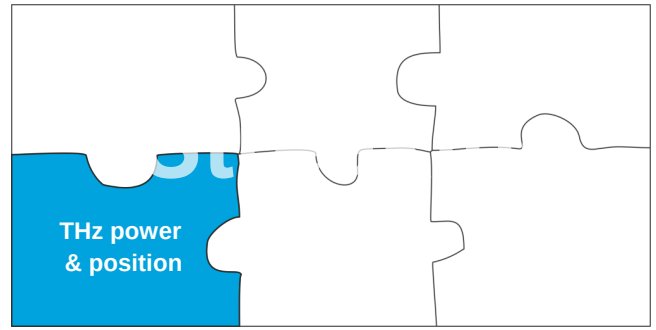
Temporal overlap

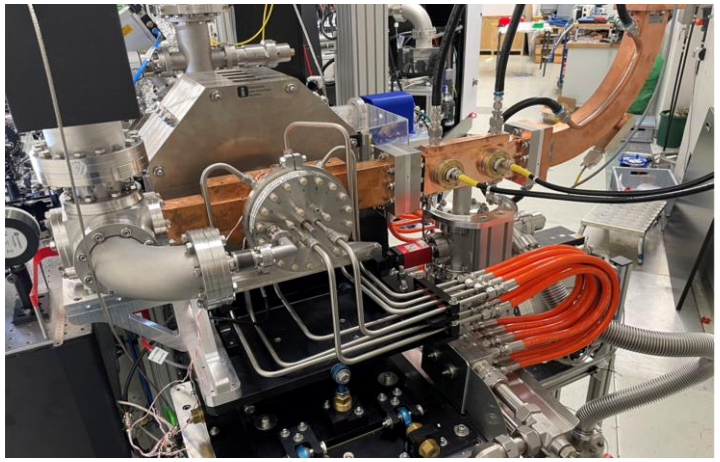
Resonator setup & design

Data evaluation

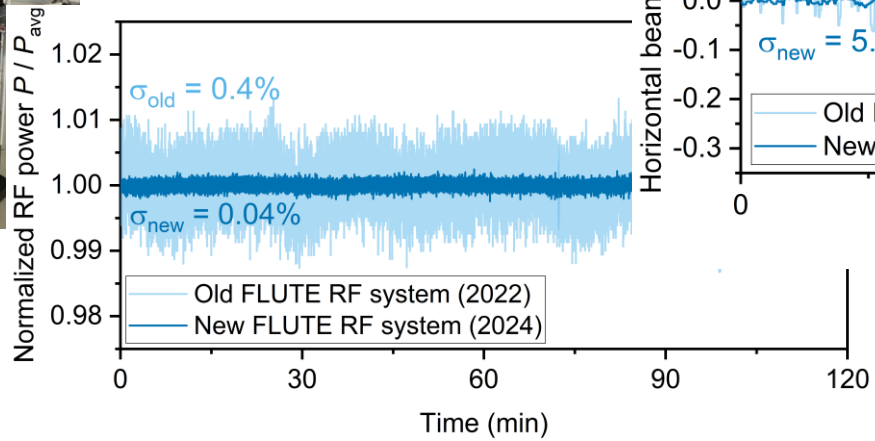
Puzzle piece: Machine stability

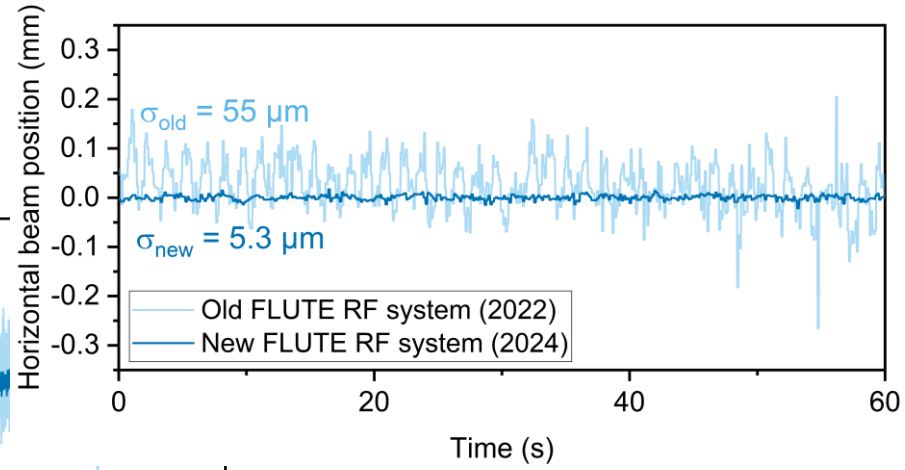


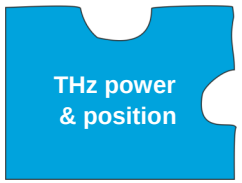


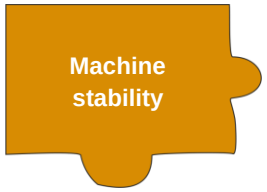


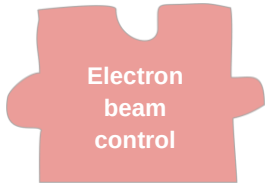




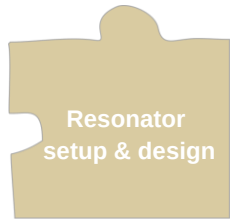






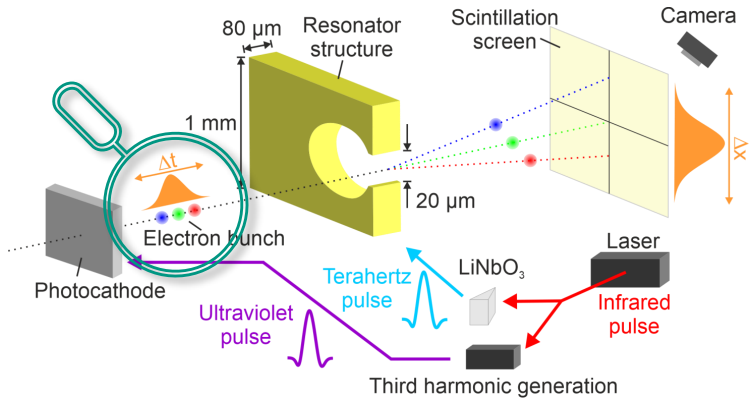


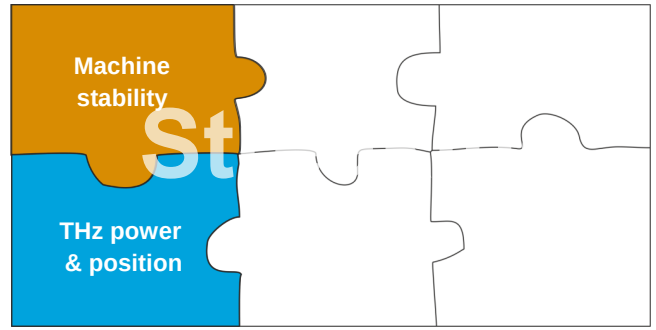


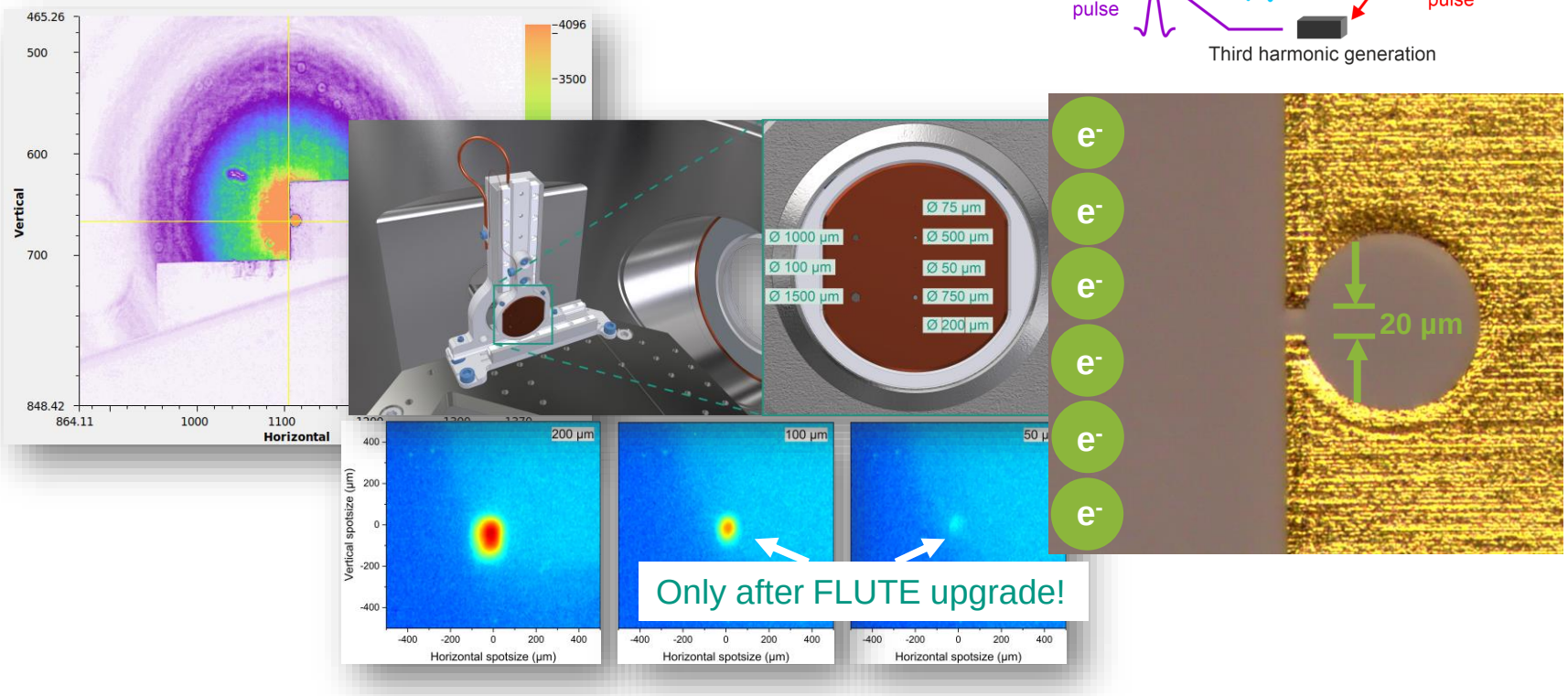


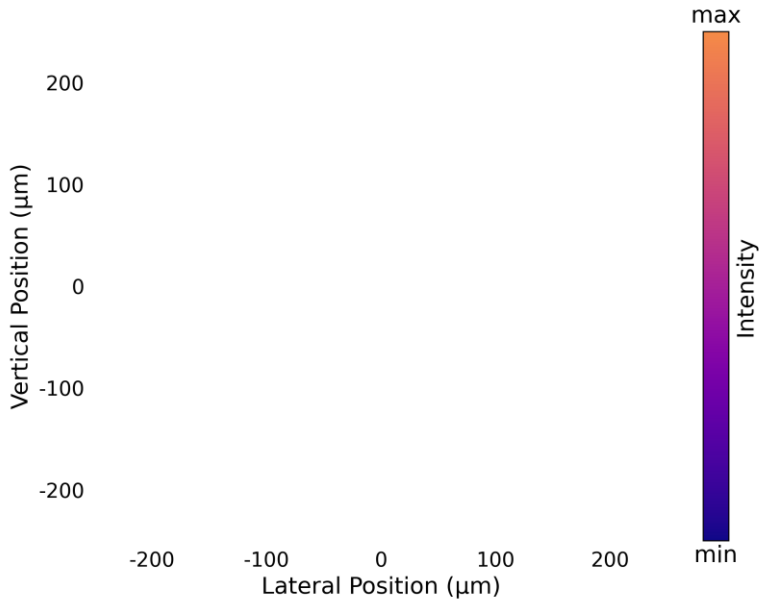


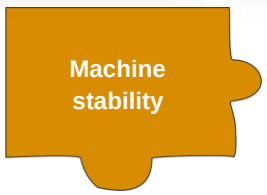
Puzzle piece: Electron beam control

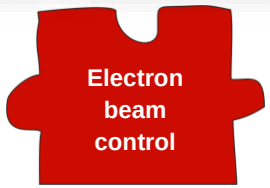




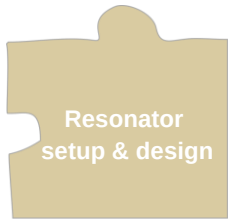






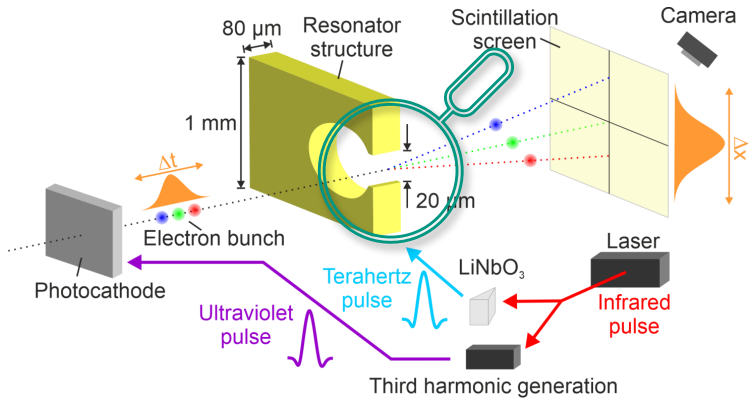


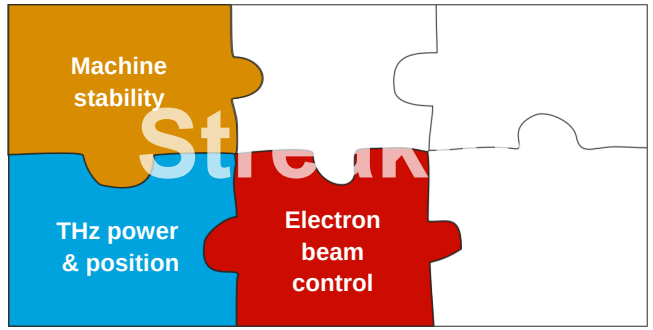


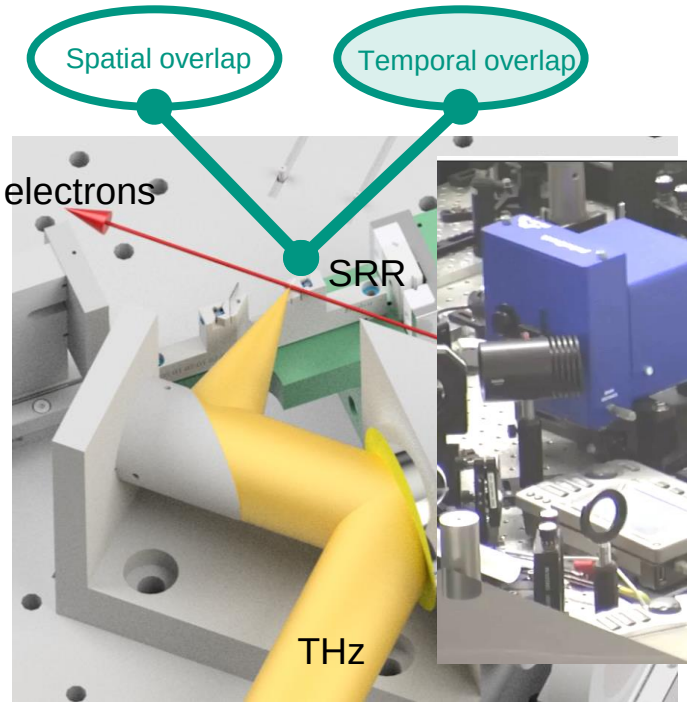


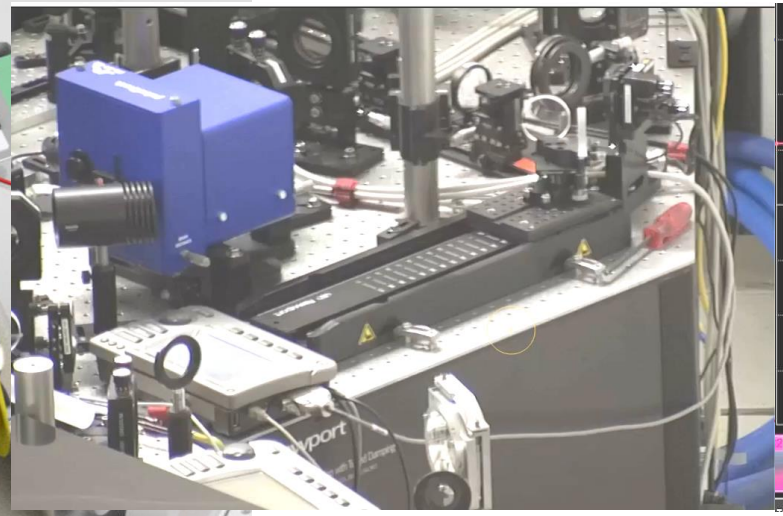


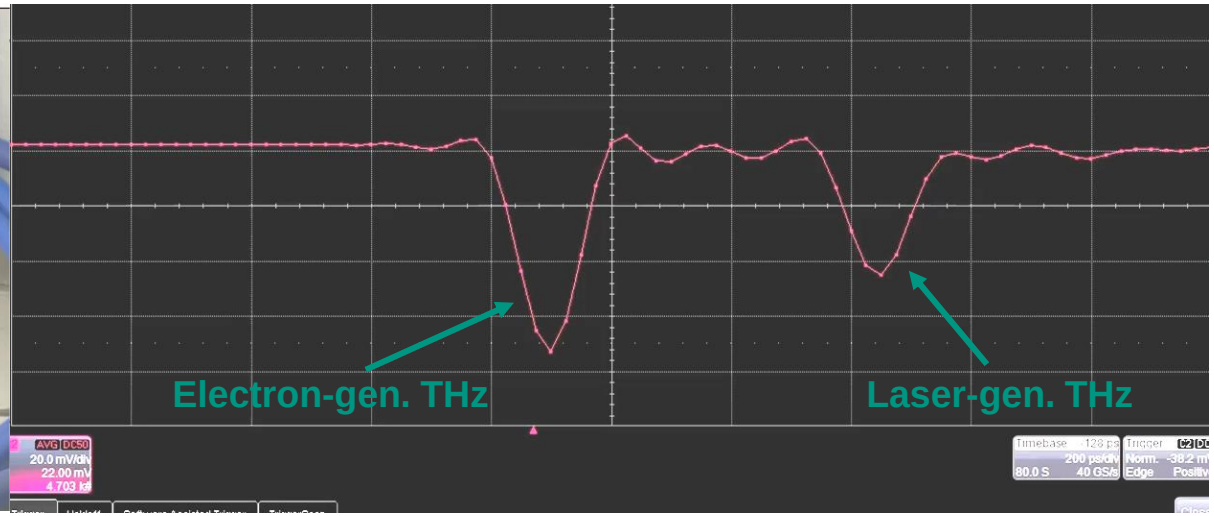
Puzzle piece: Temporal overlap

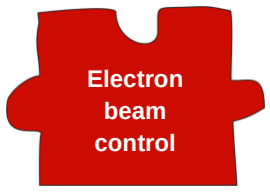




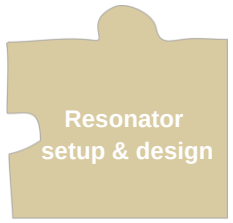






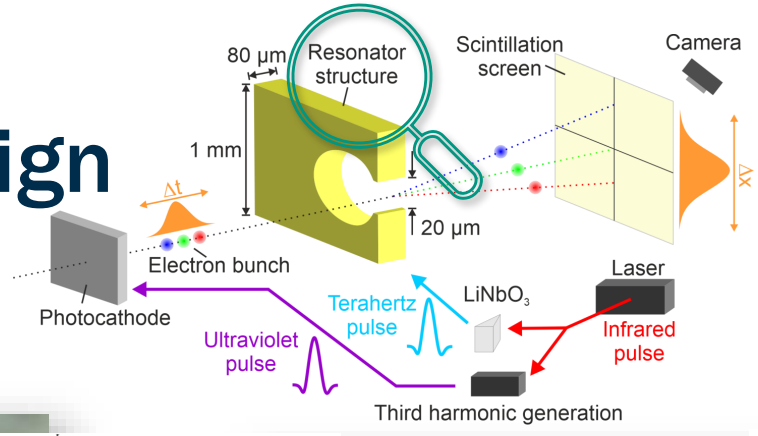


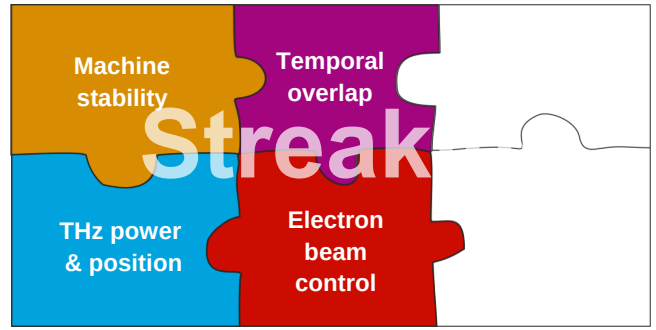


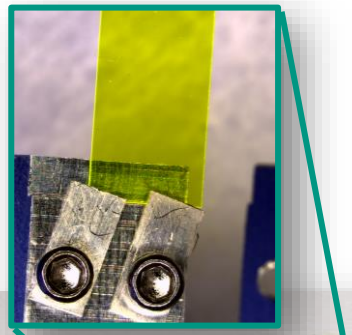


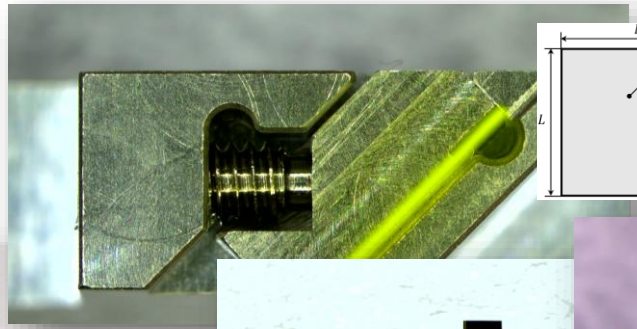


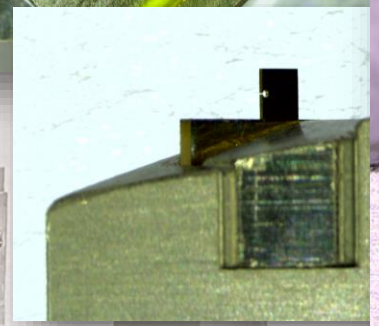
Puzzle piece: Resonator setup & design

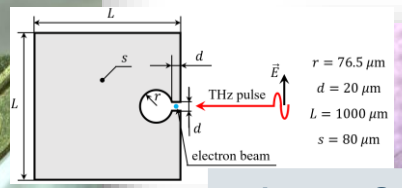






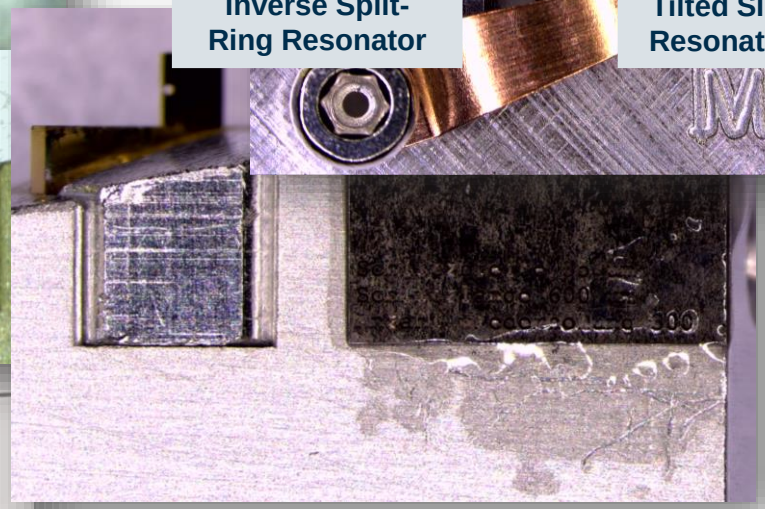


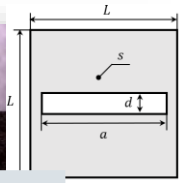




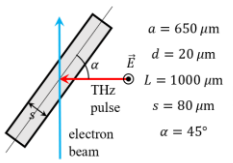


Inverse Split-Ring Resonator

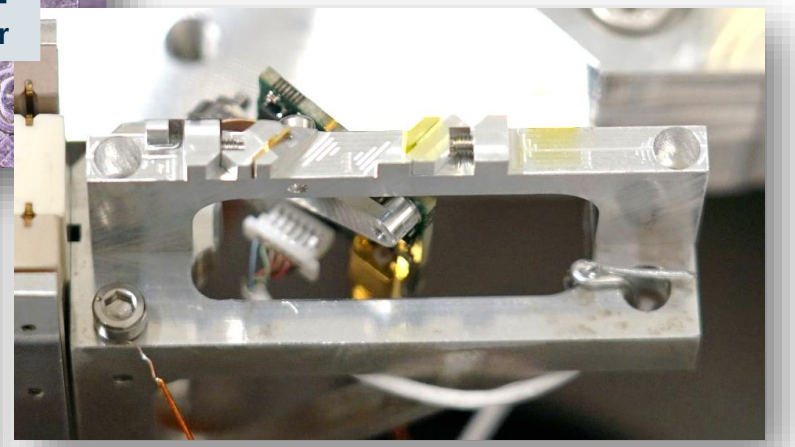




Tilted Slit-Resonator



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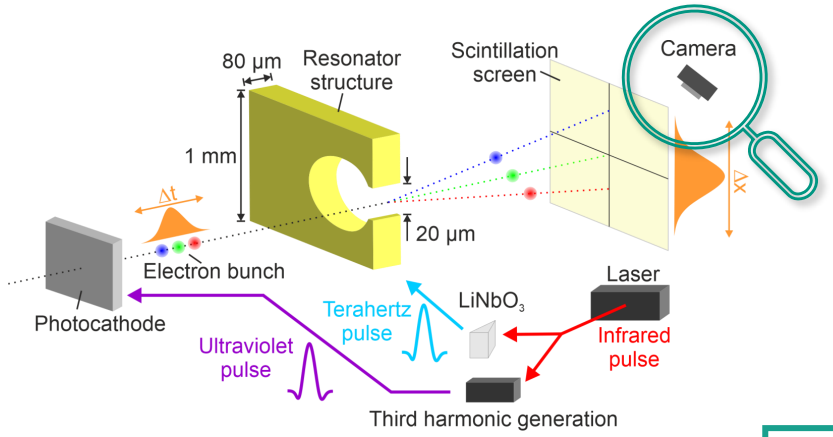


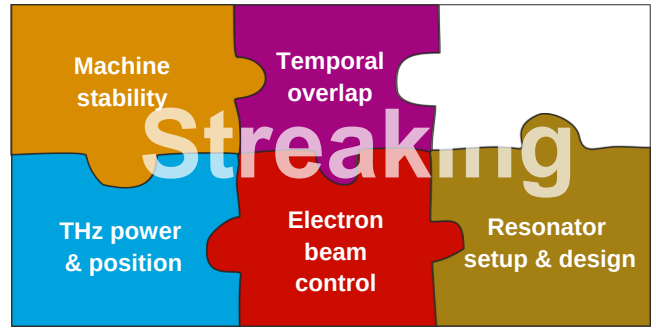
Temporal overlap

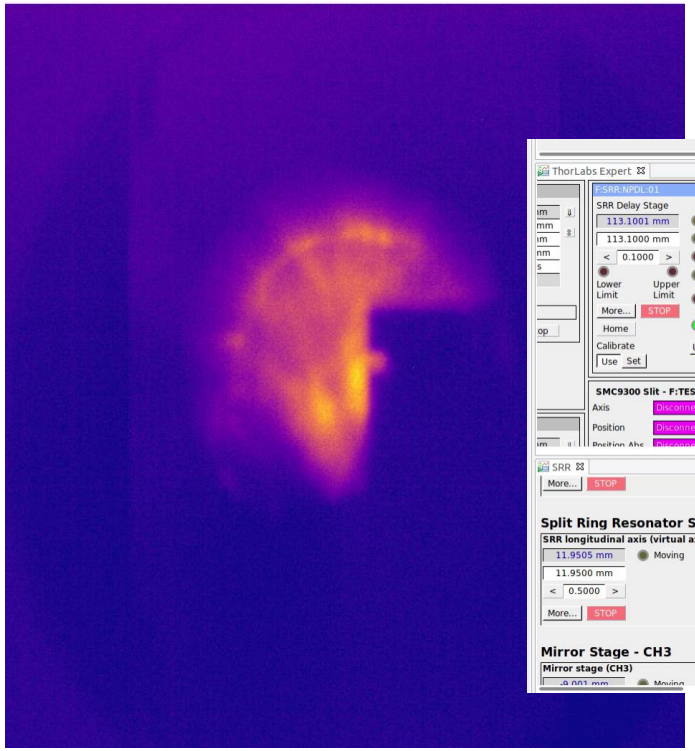
Resonator setup & design

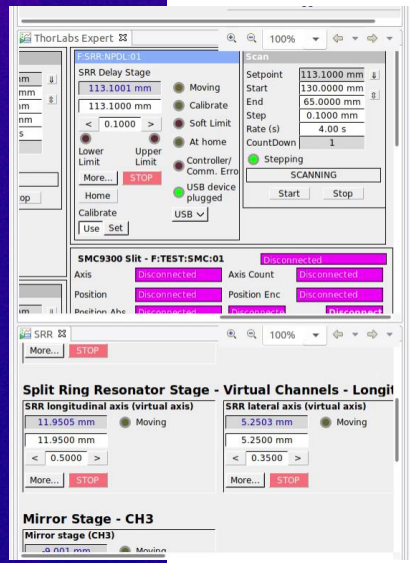
Data evaluation

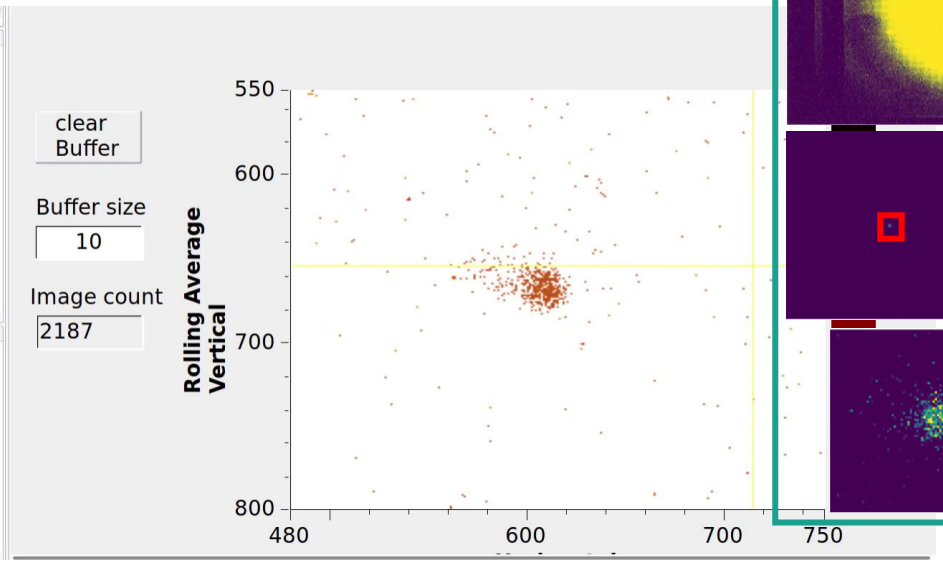
Puzzle piece: Data evaluation

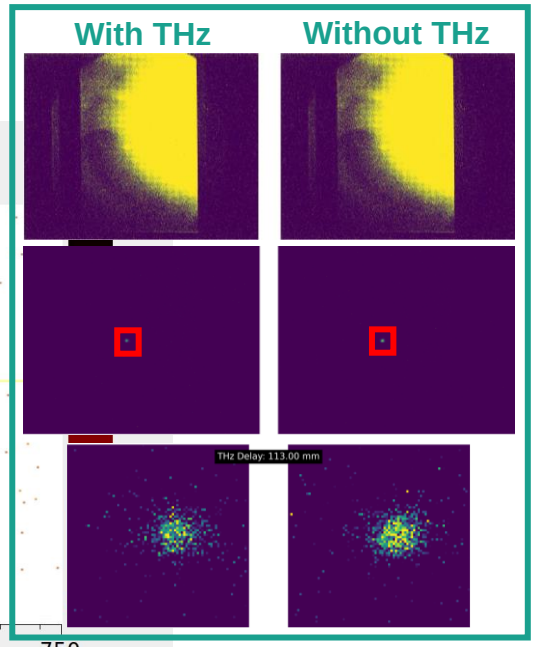










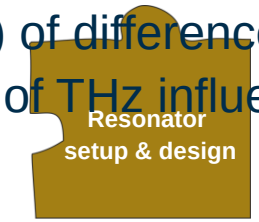


w/ BG

w/o BG

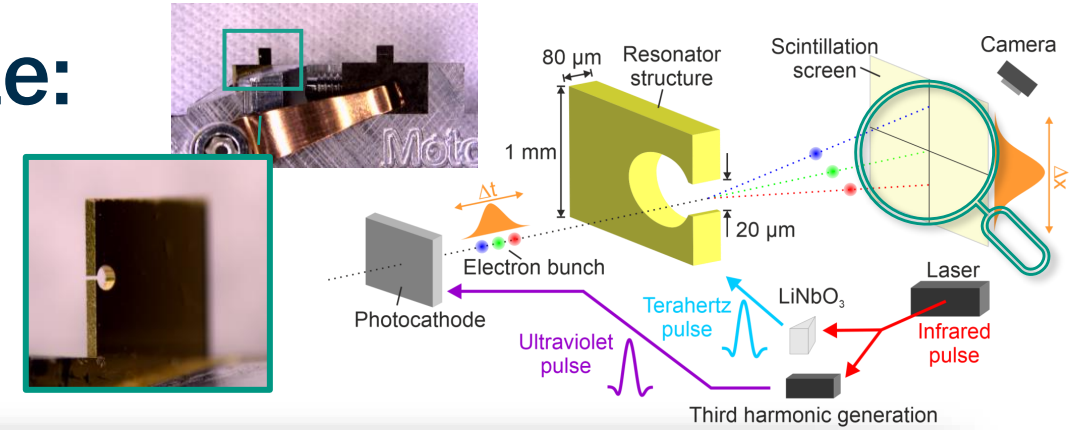
ROI

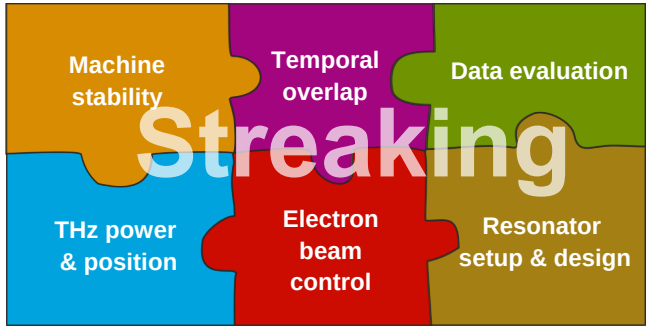
- Calculating mean absolute deviation (MAD) of difference image
- High MAD ➡ Low Similarity ➡ Indication of THz influence

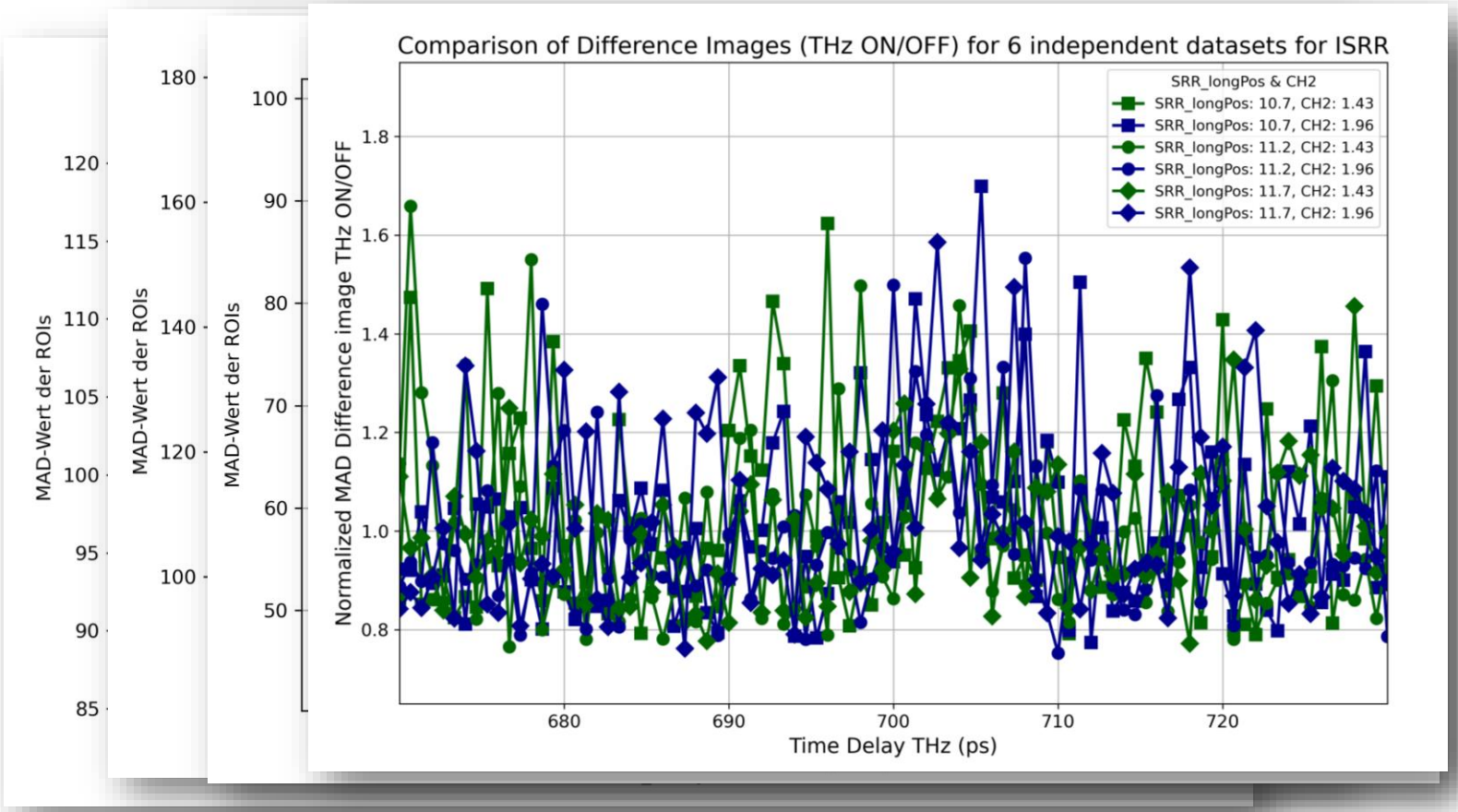


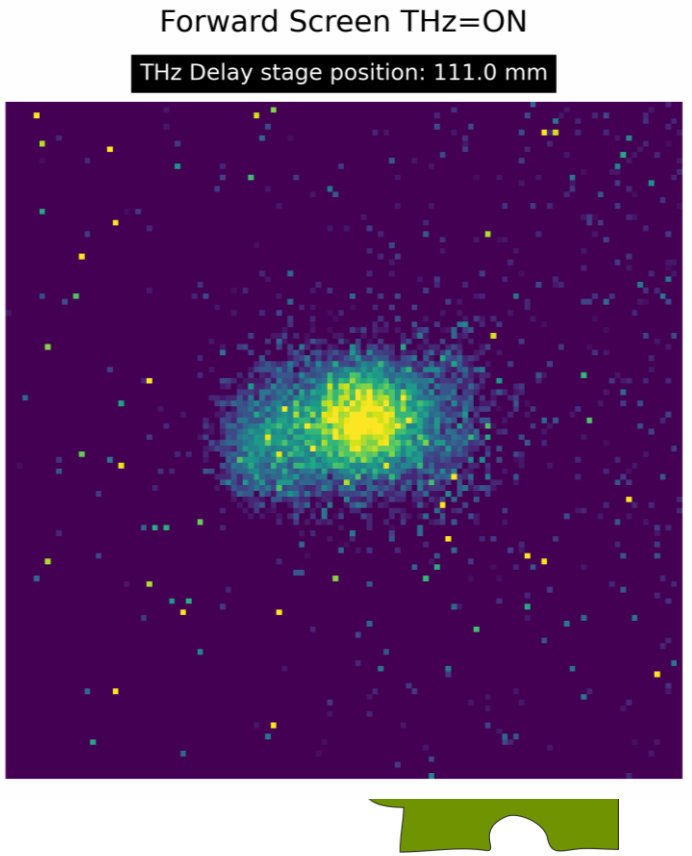


Puzzle complete: Streaking!

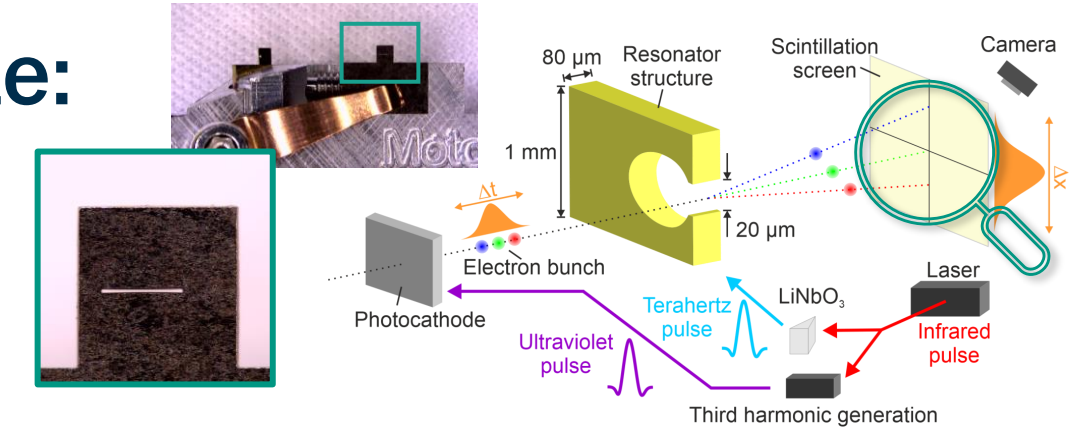


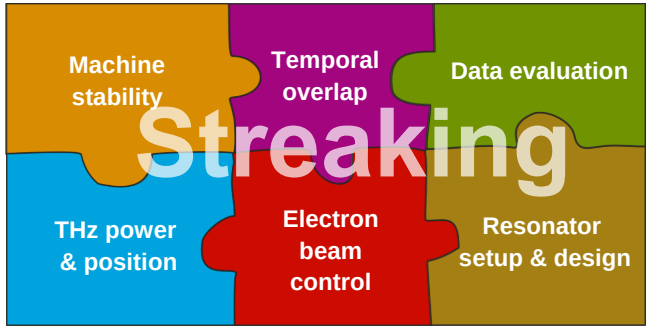


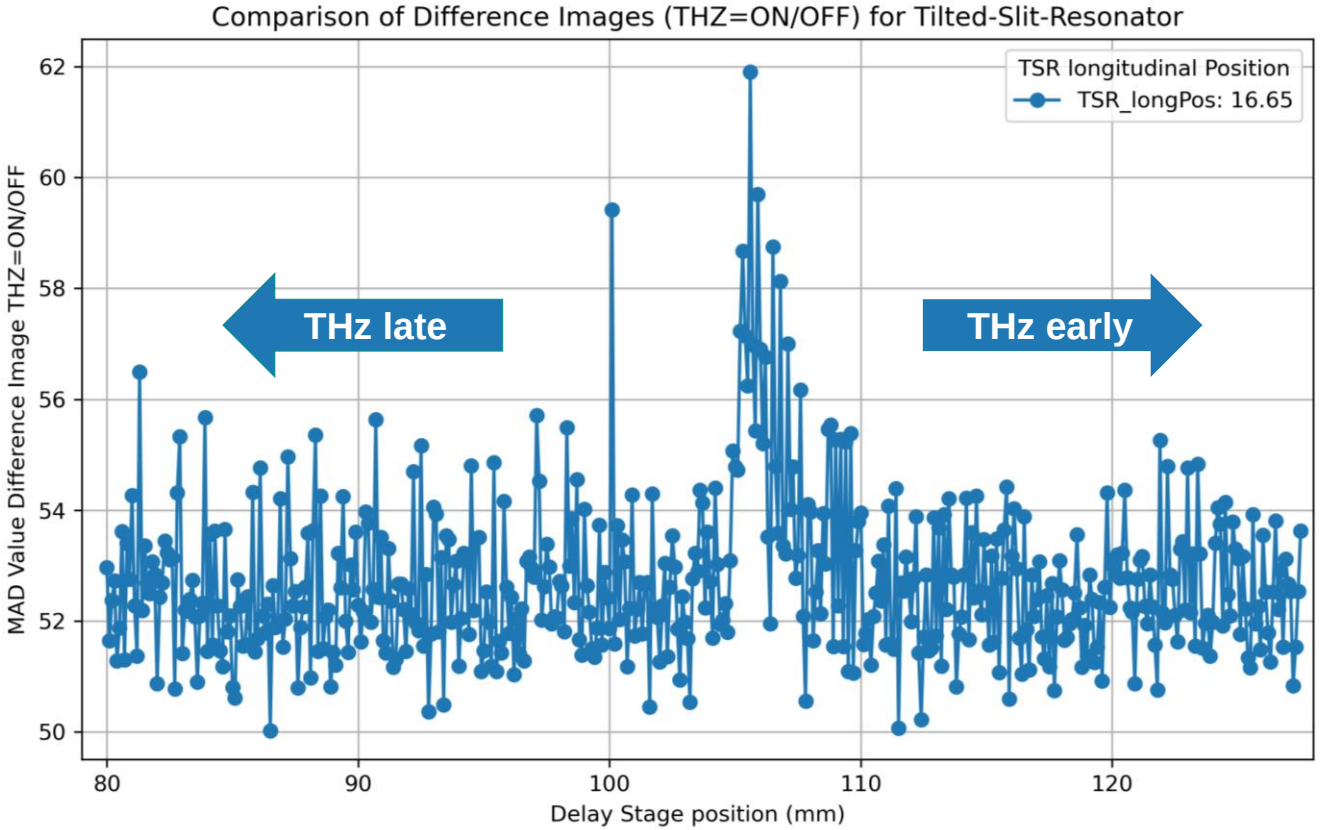




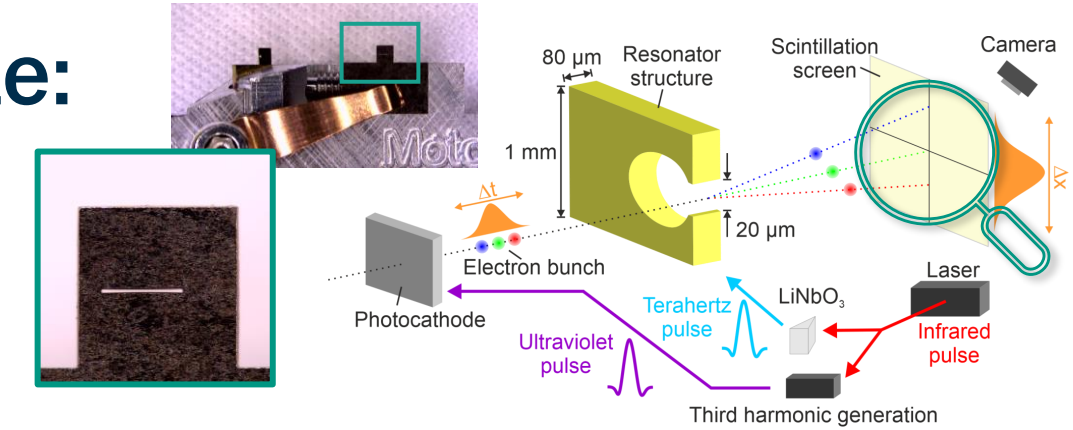
Puzzle complete: Streaking!

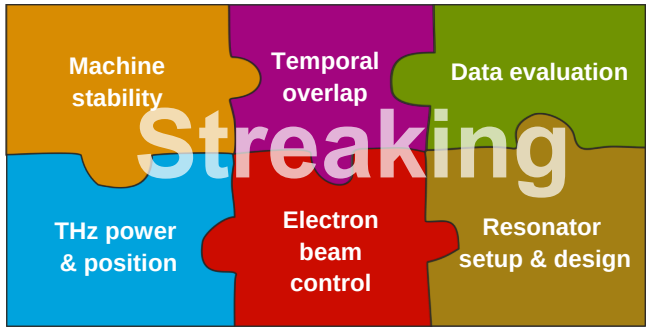


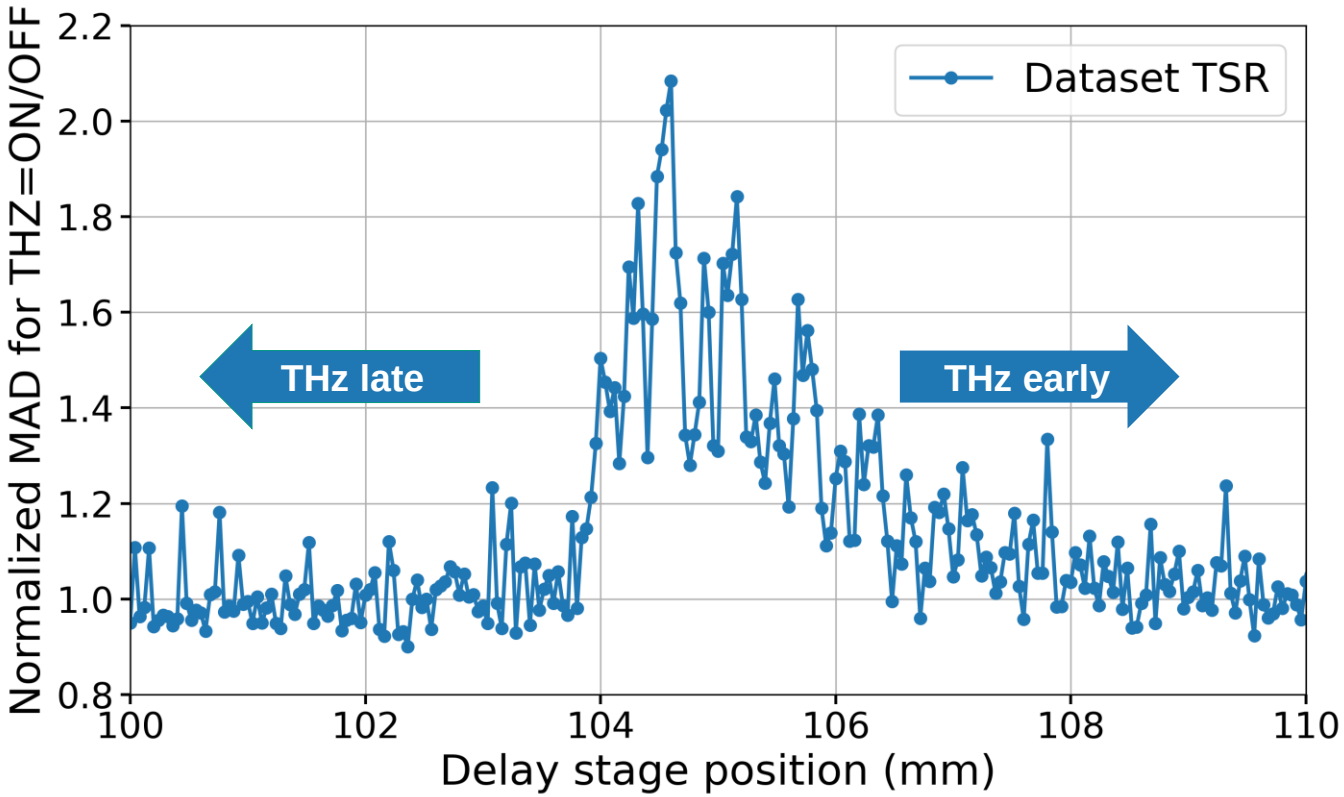




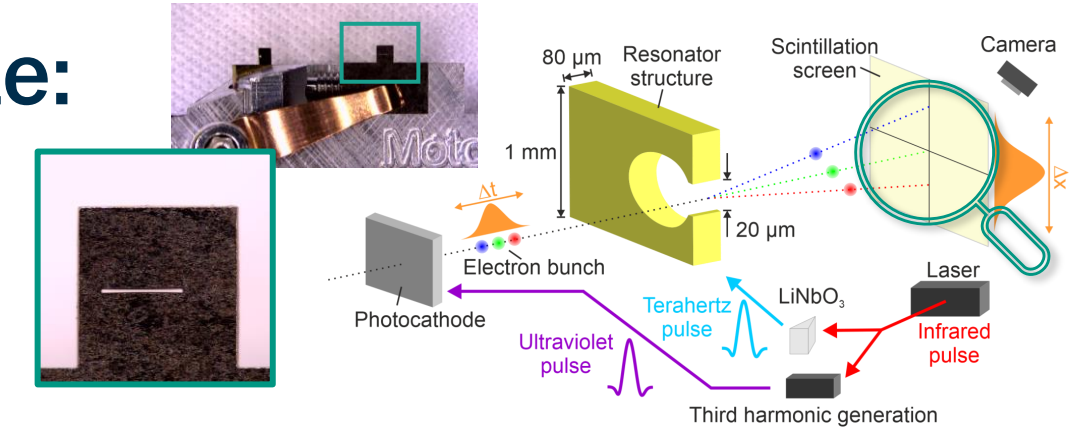
Puzzle complete: Streaking!

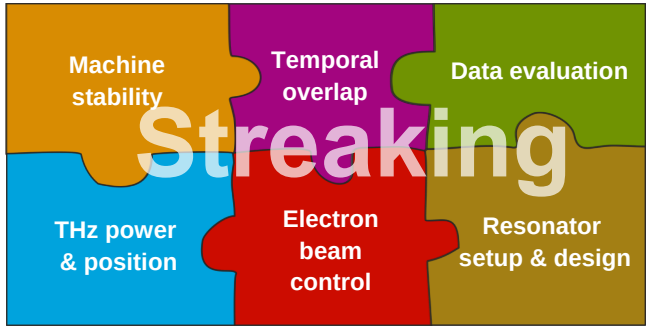


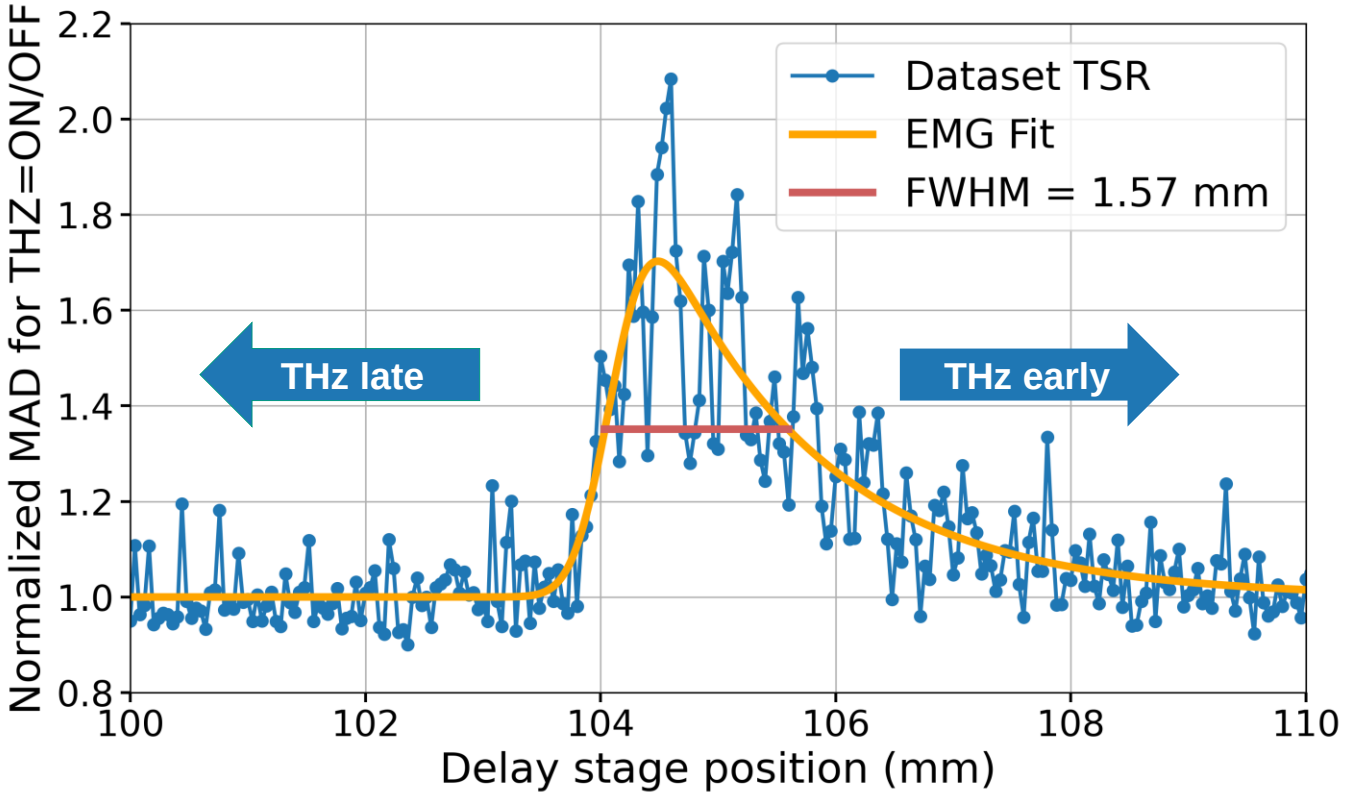


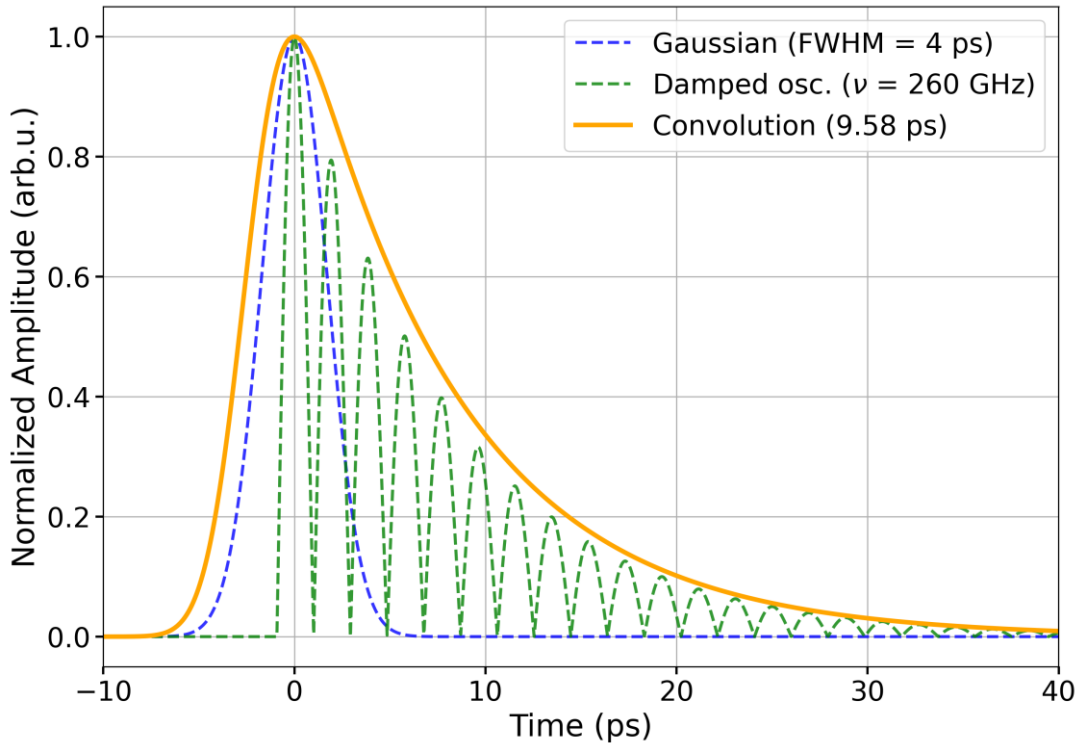


Puzzle complete: Streaking!

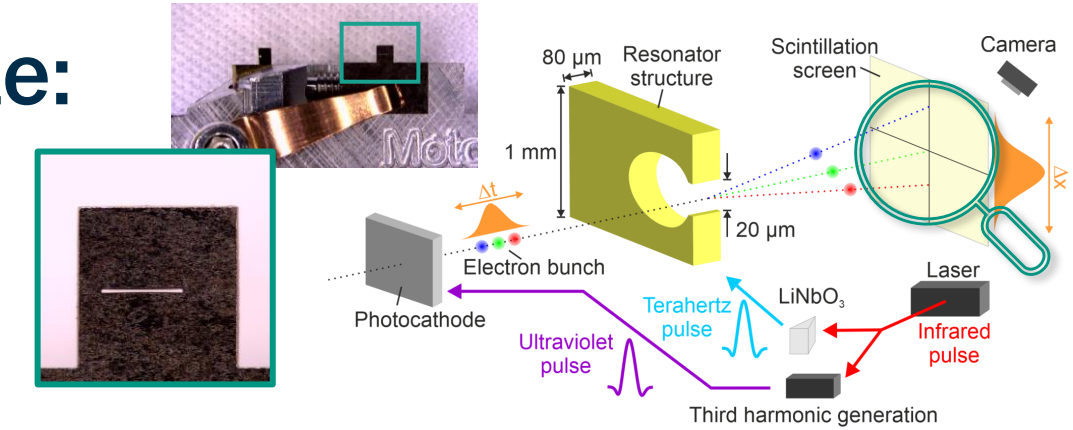


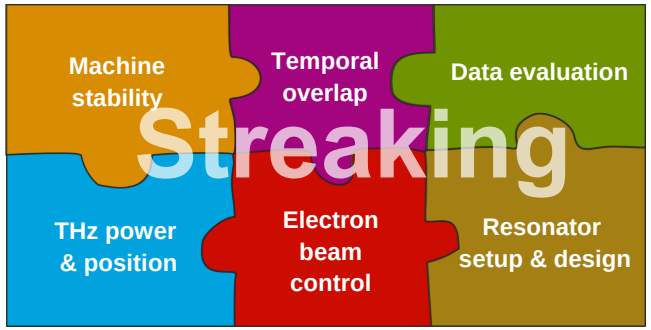


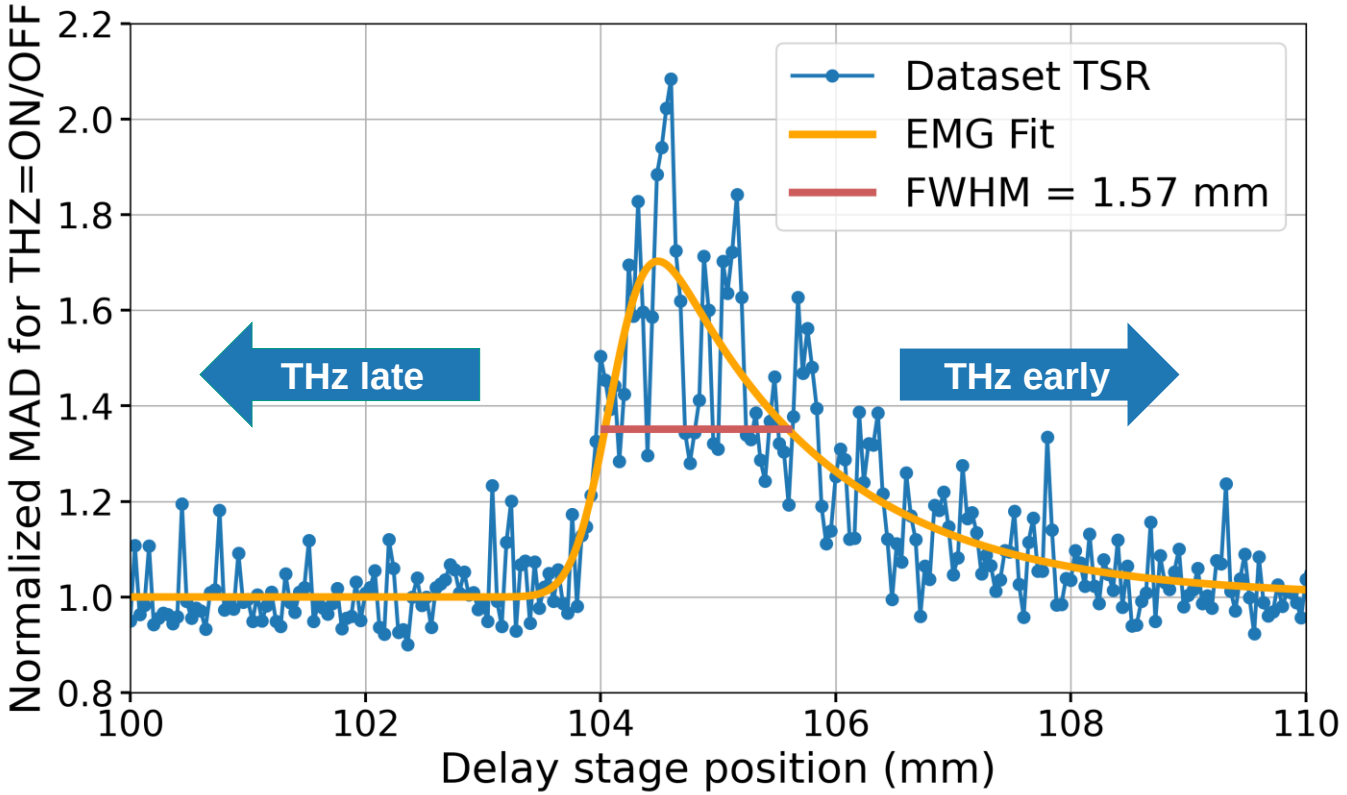


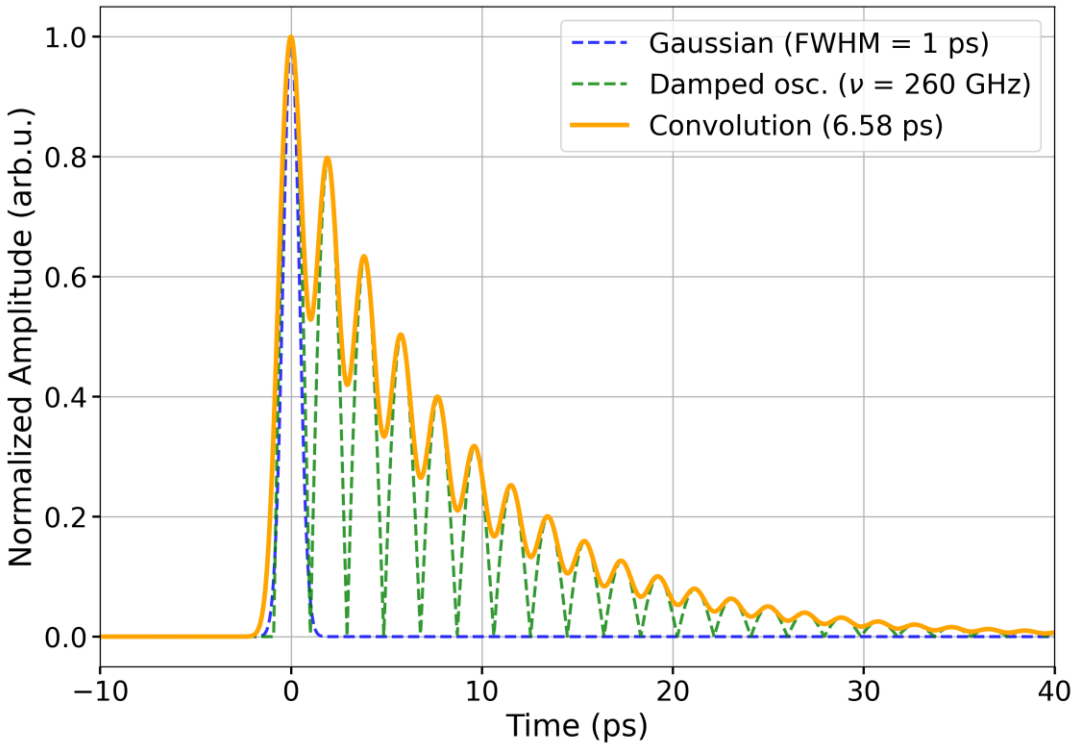


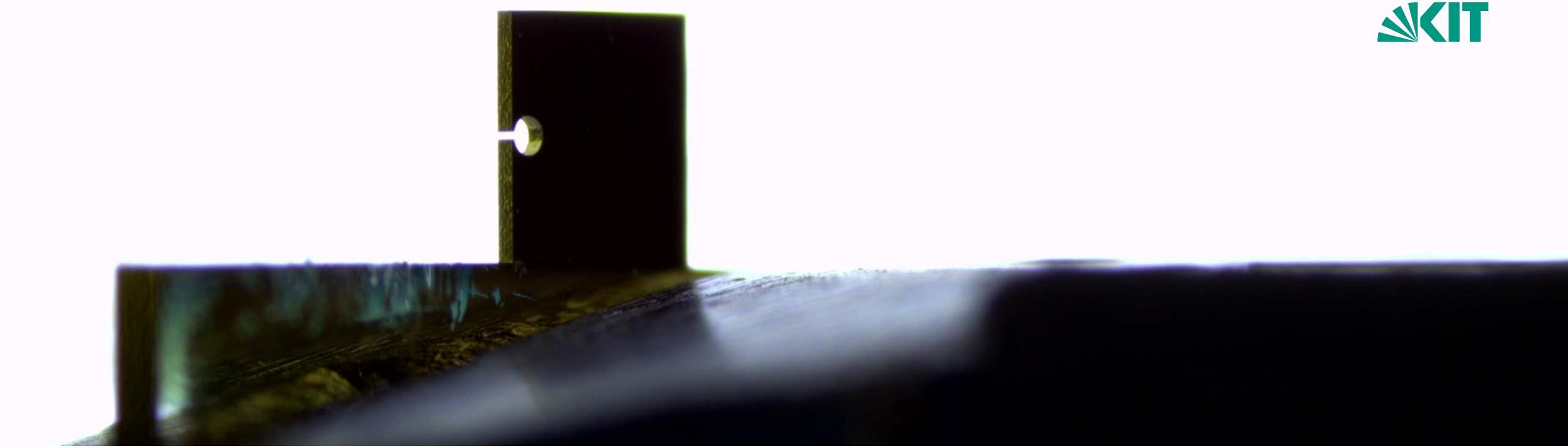
Puzzle complete: Streaking!









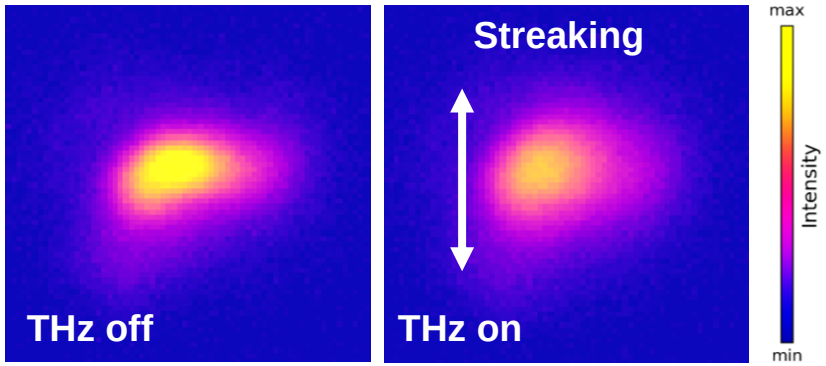


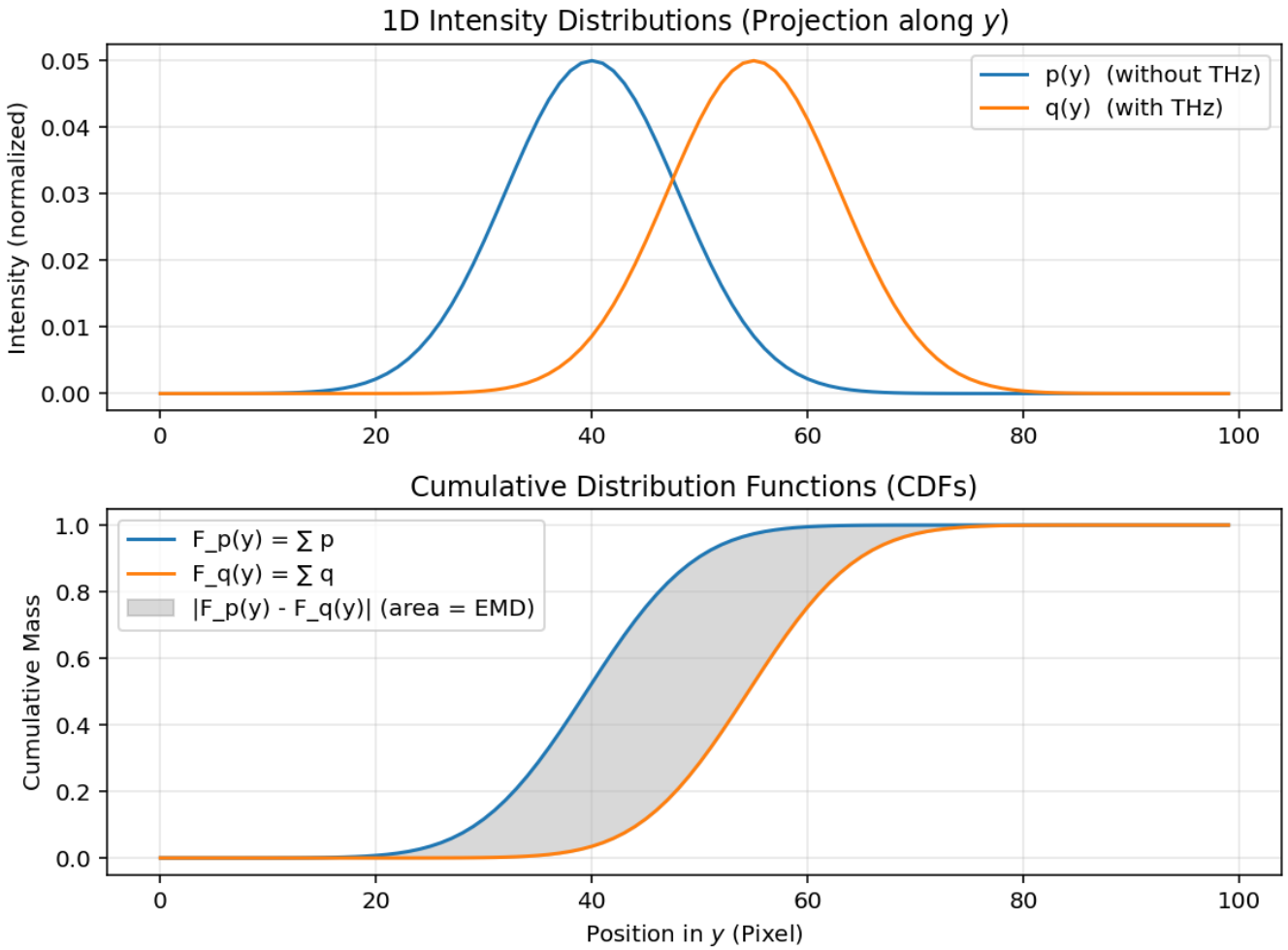
Streaking Analysis–
(Very quick) Wasserstein Metric

Matthias Nabinger
on behalf of the Compact-TDS collaboration

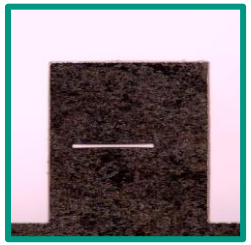
Wasserstein Metric

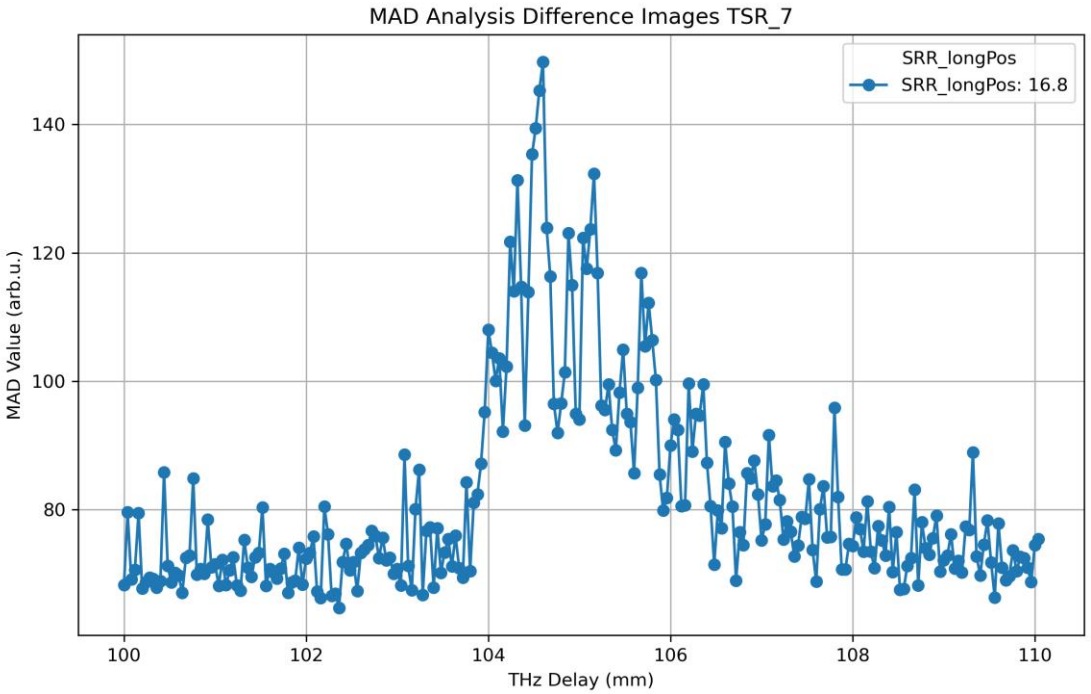
- **1D Idea: CDF comparison**
  Measures how much “mass” (intensity) must be moved to transform one distribution into another
- **Physical Intuition**
  2 piles of sand → minimum work to reshape one into the other
- **2D extension**
  → Transport field where and how much shifting occurs

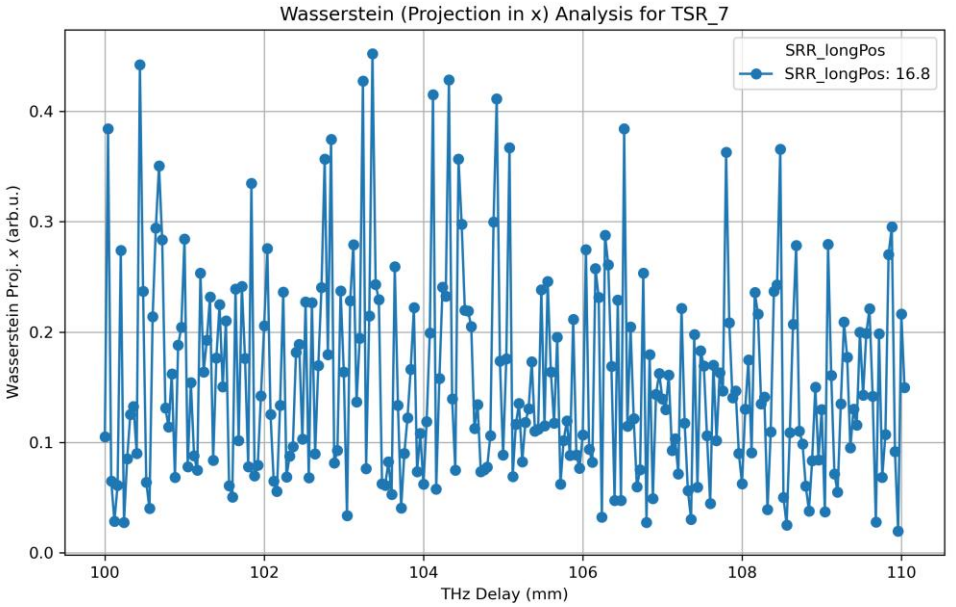


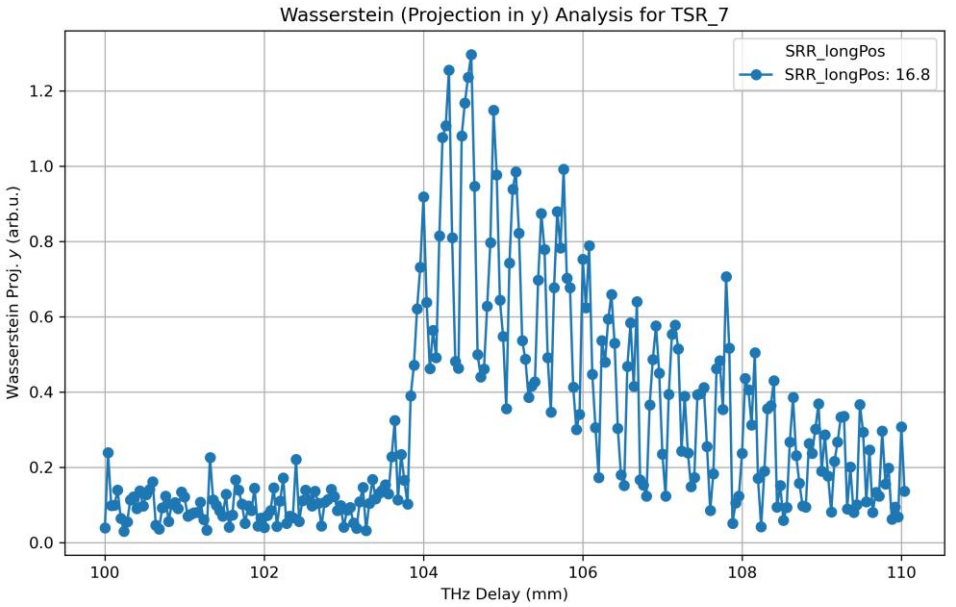


Comparison MAD – 1D

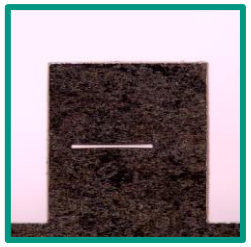


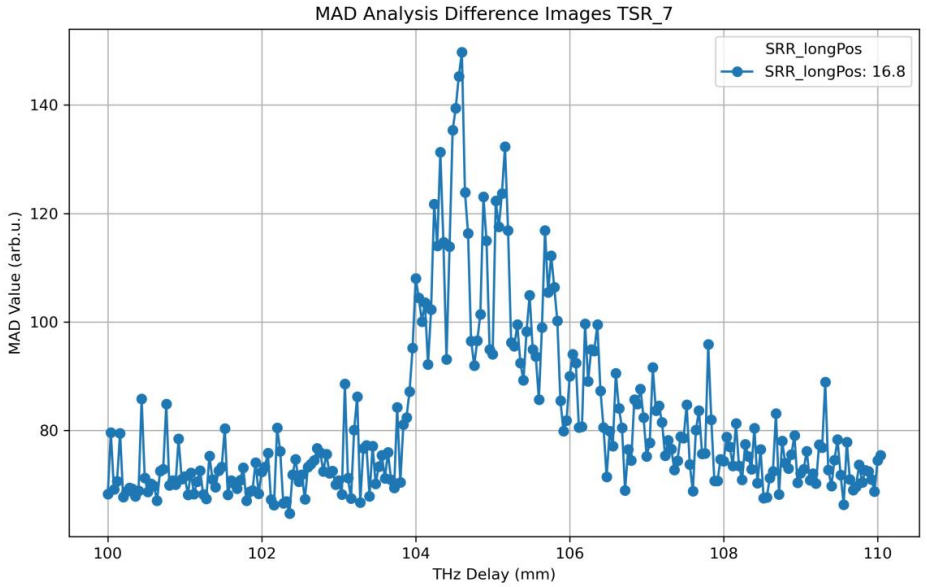


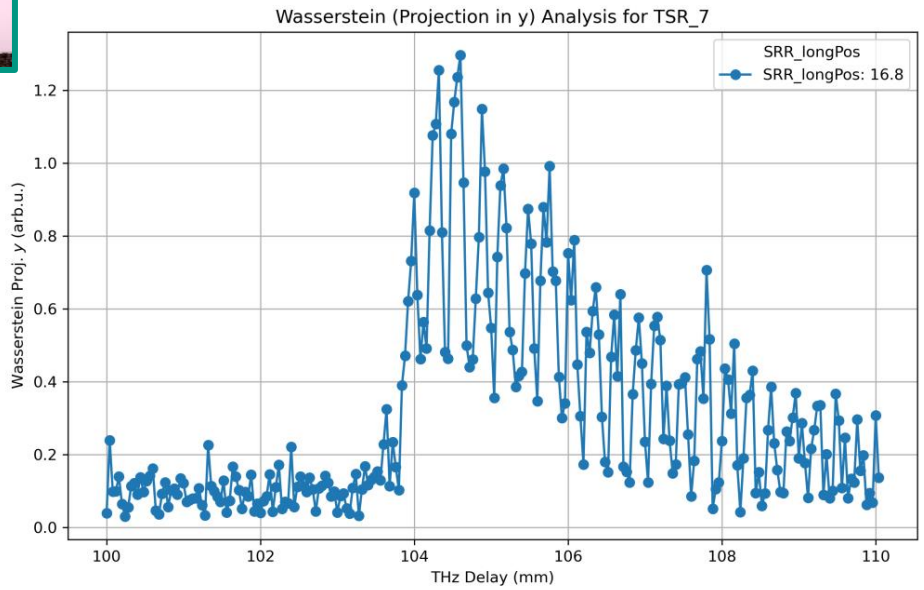


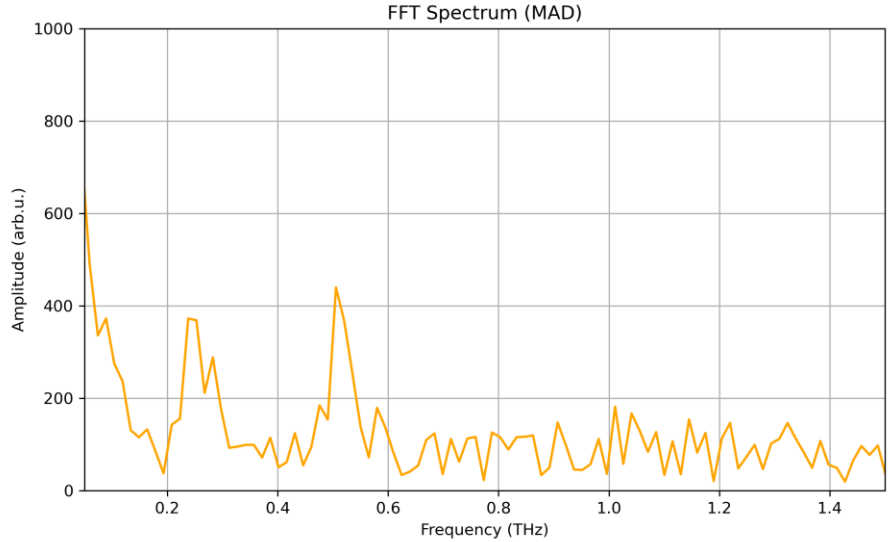


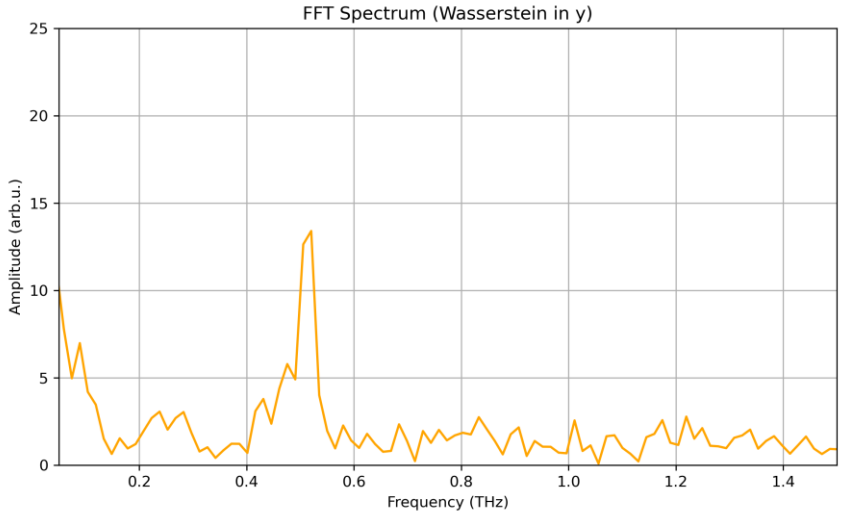
Comparison MAD – 1D



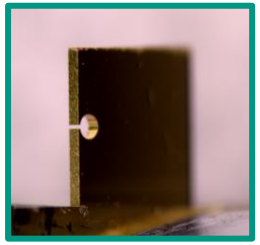


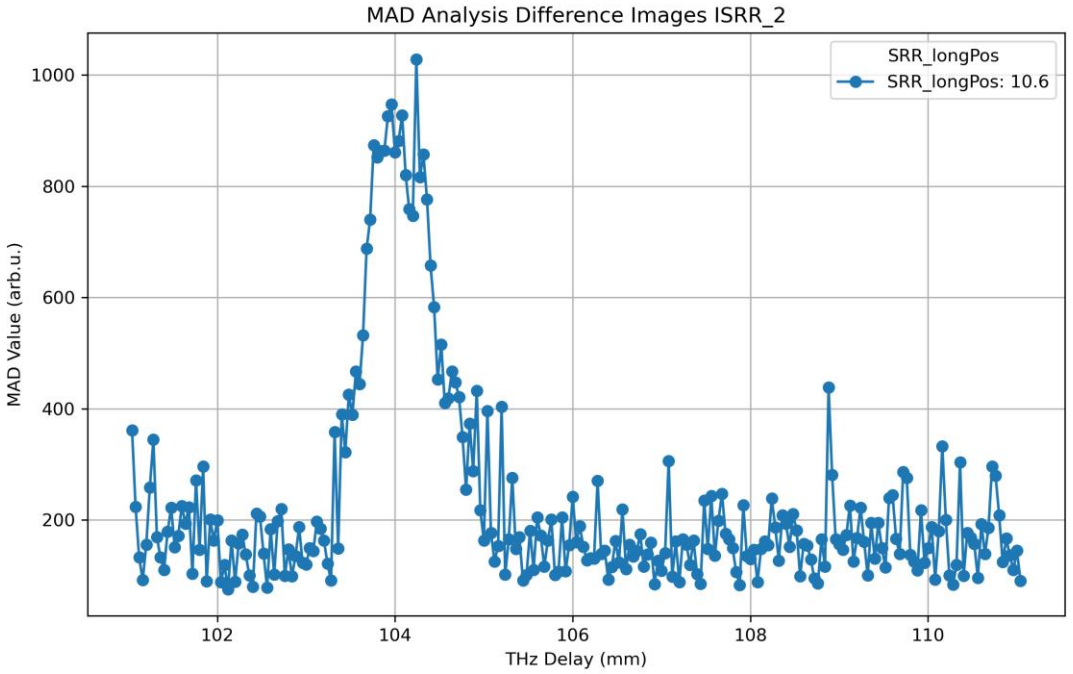


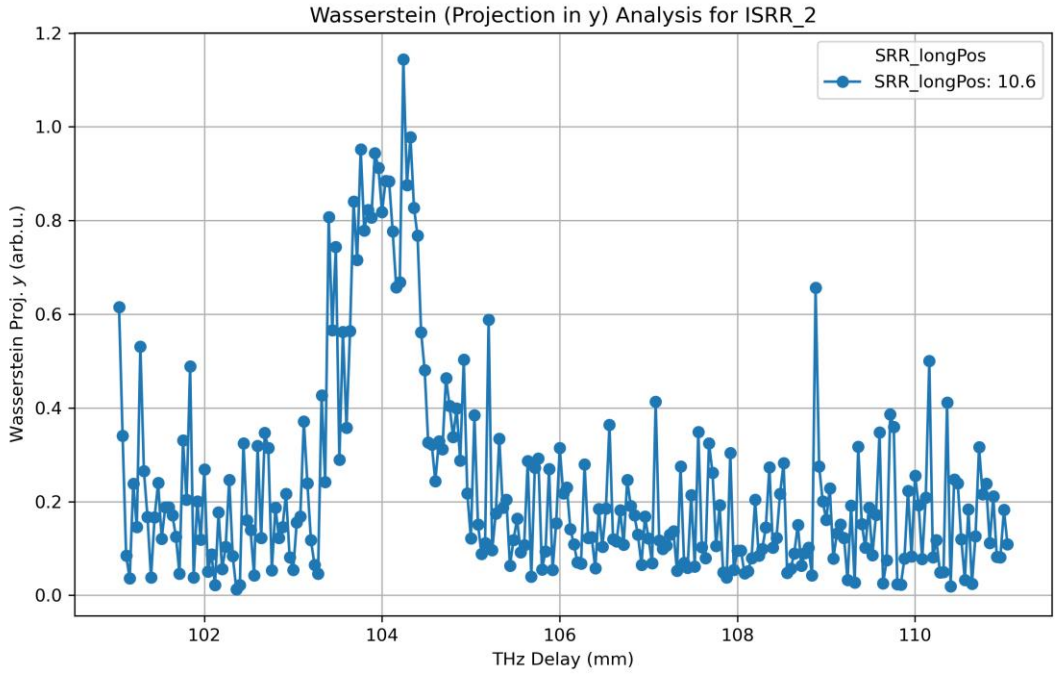




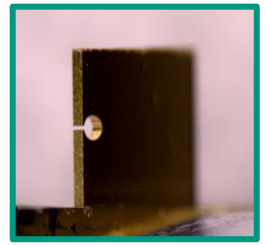
Comparison MAD – 1D

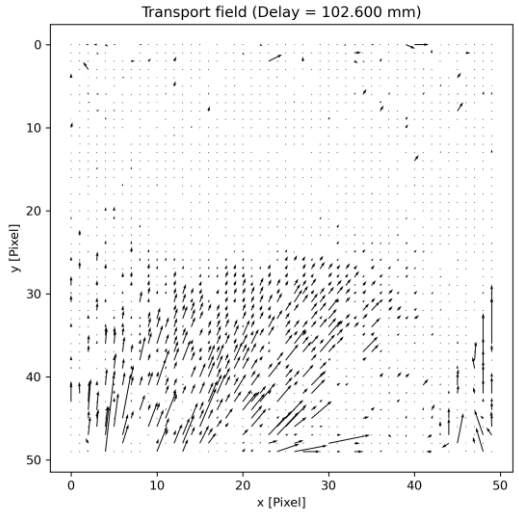


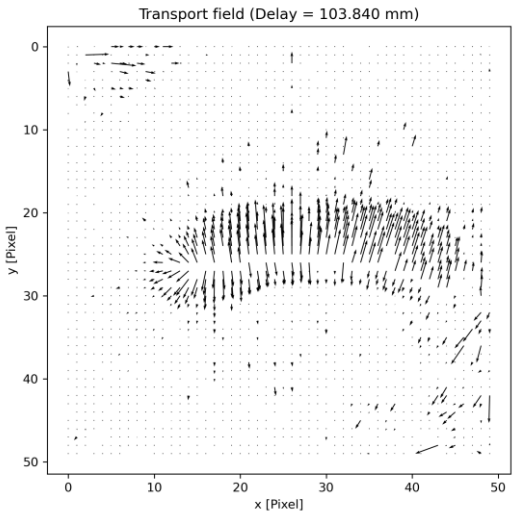


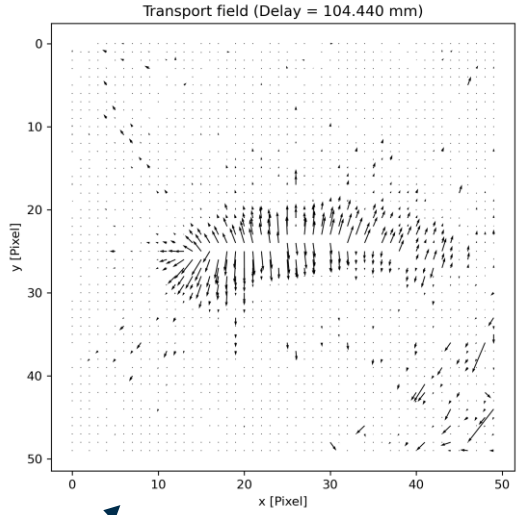


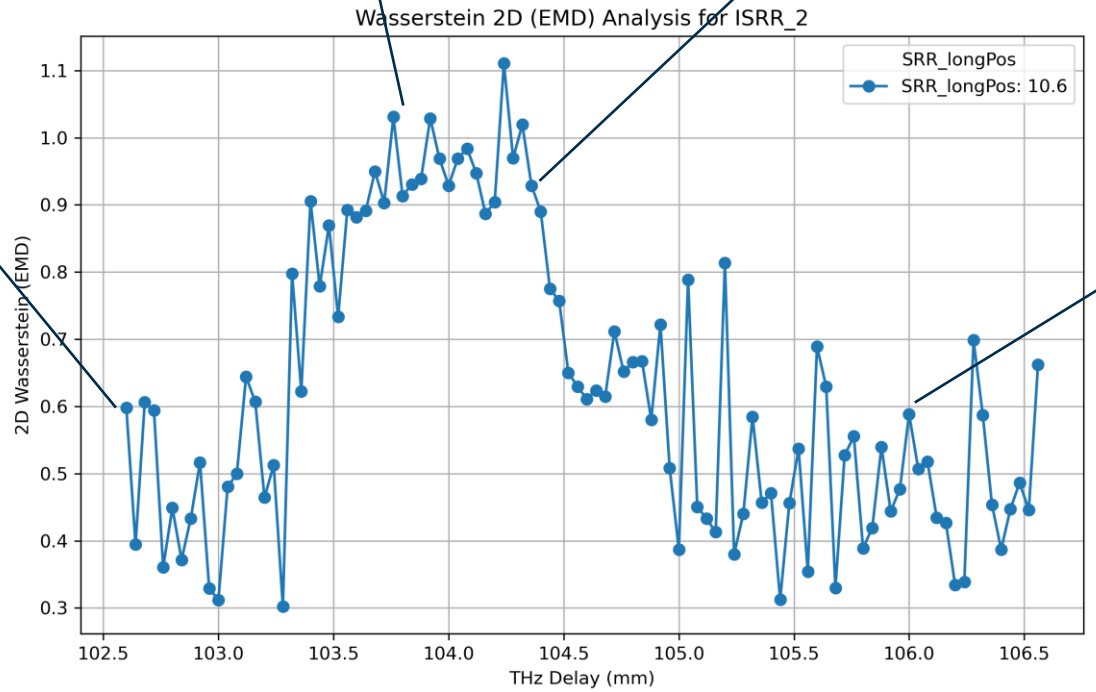
Wasserstein 2D

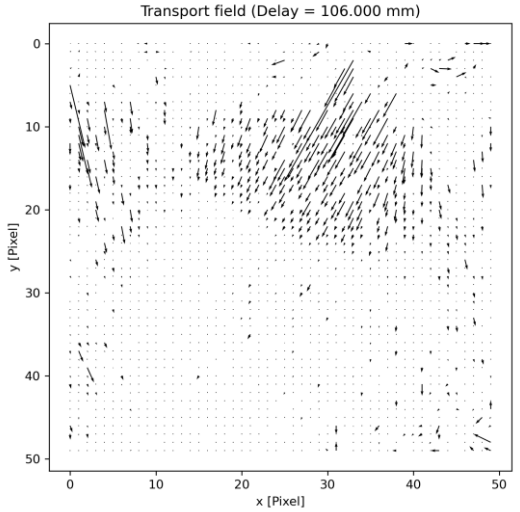




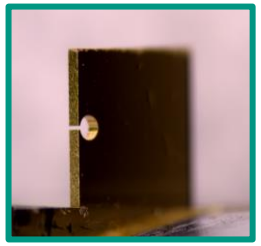


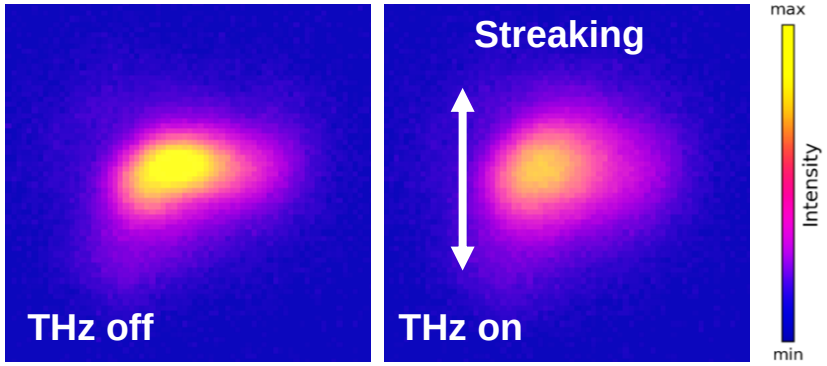


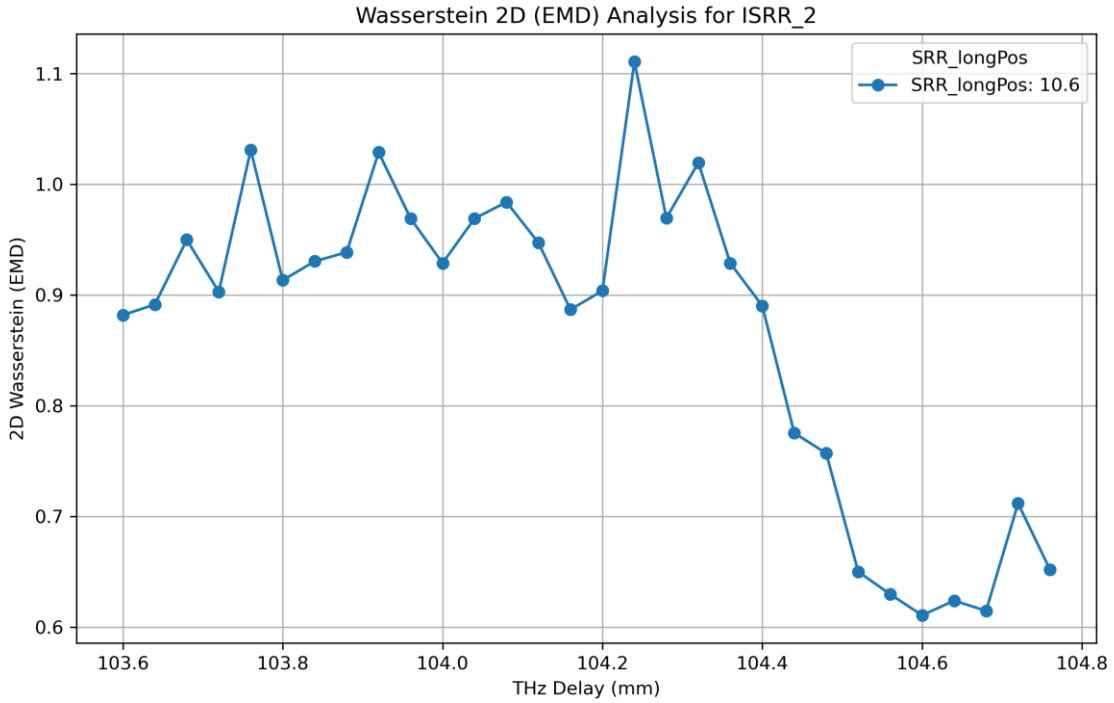


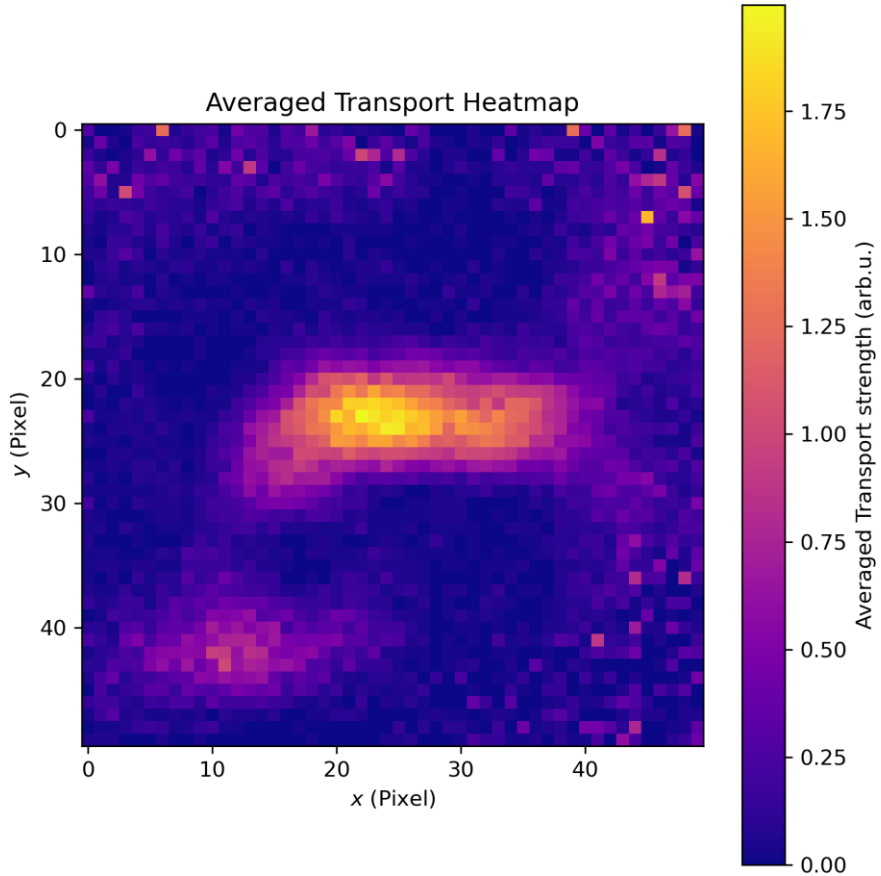


Wasserstein 2D

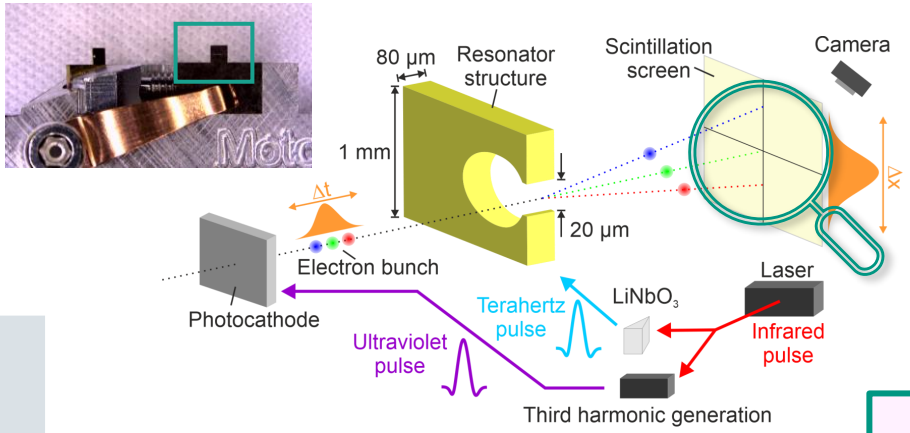


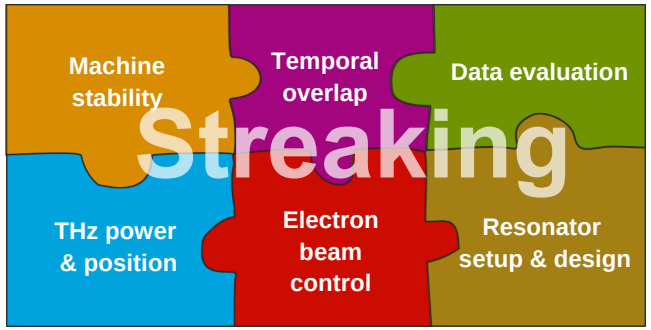


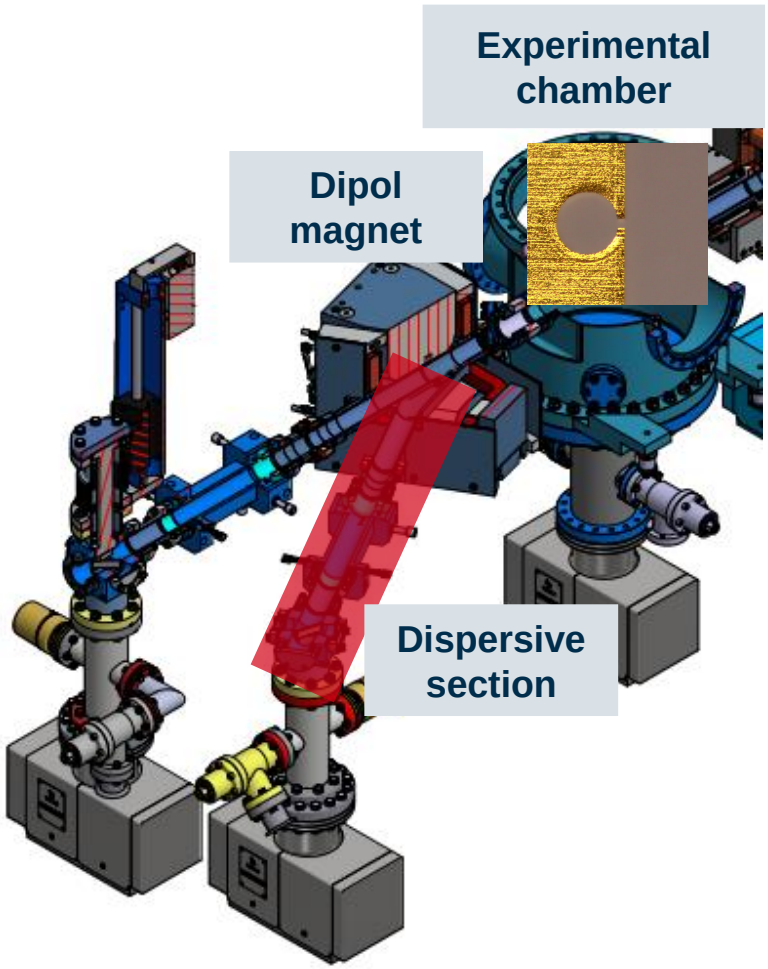




Puzzle complete: Streaking!

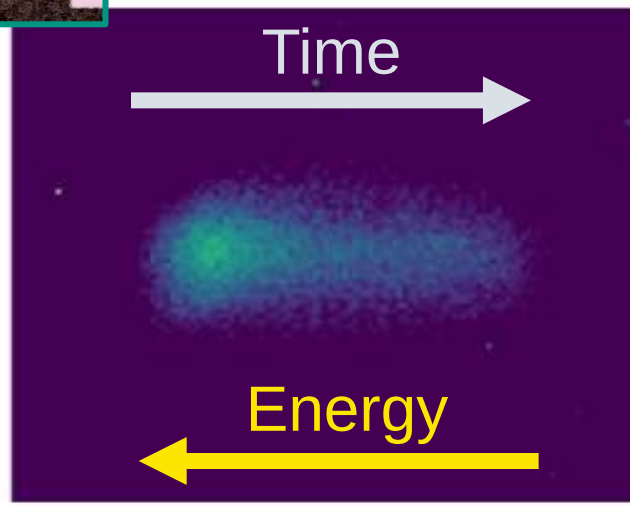




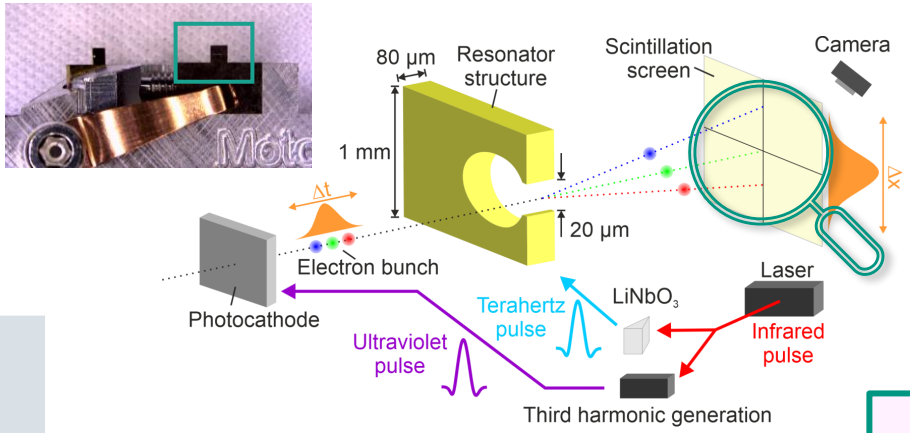


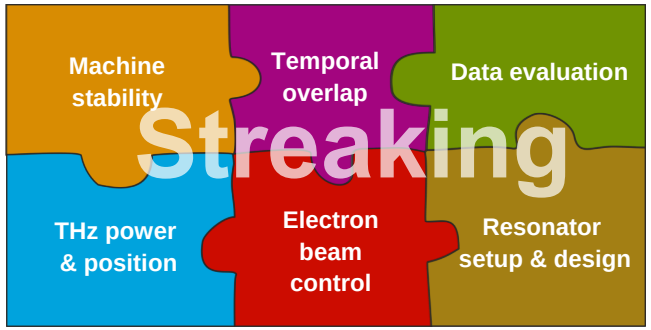


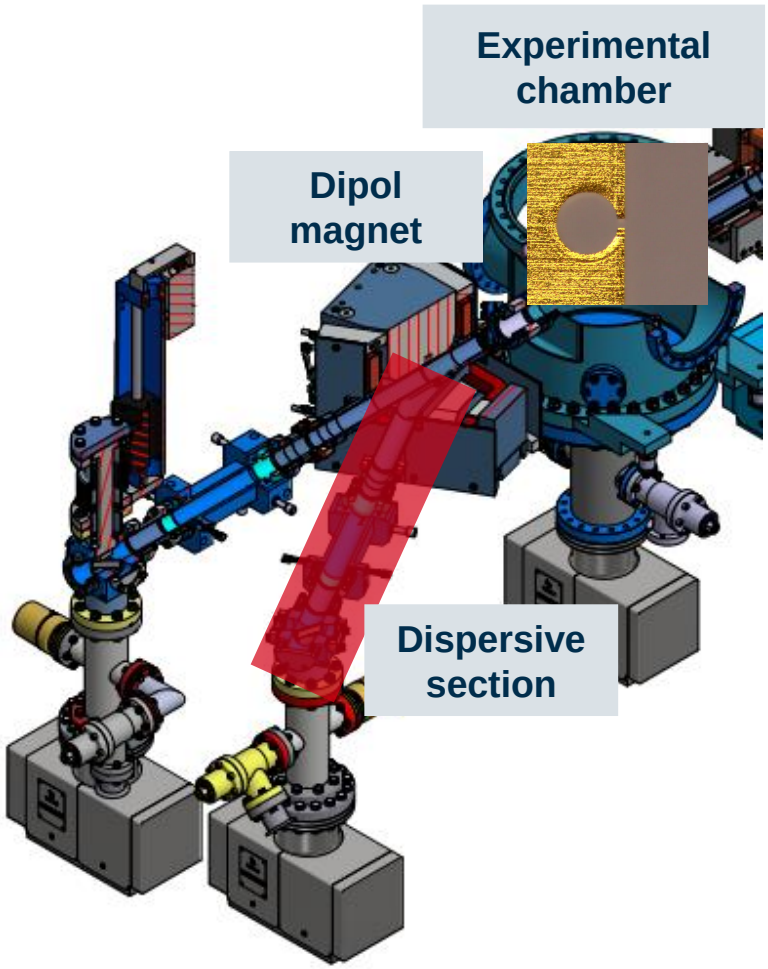


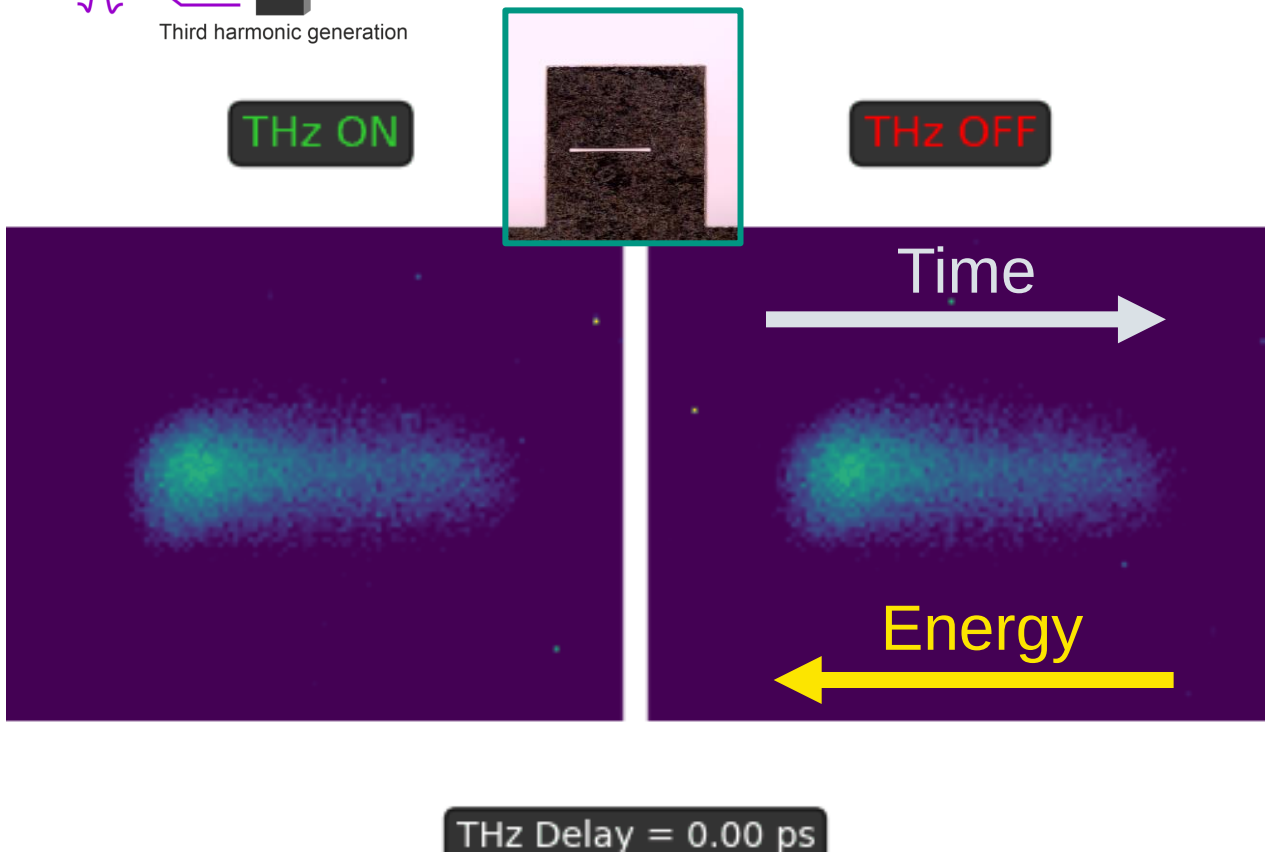


Puzzle complete: Streaking!

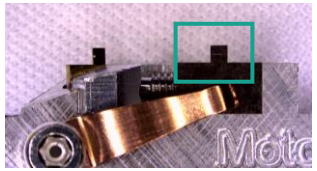


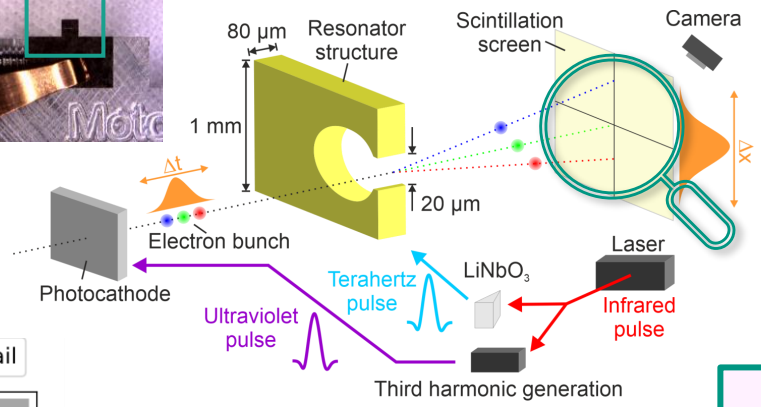


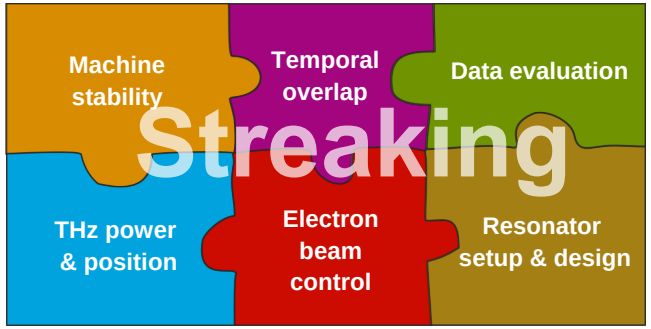


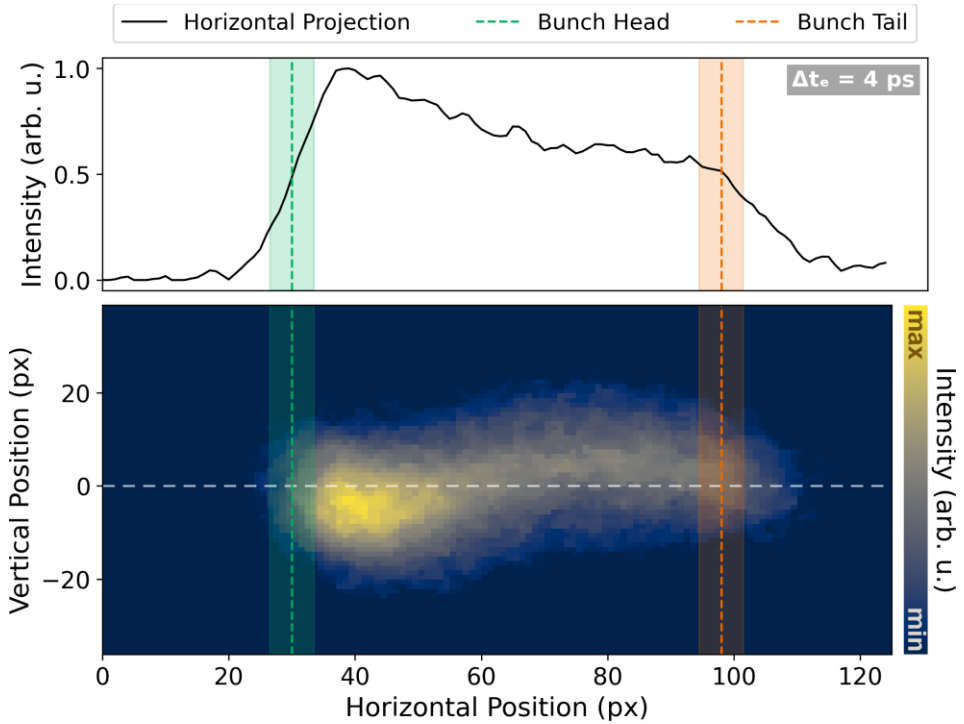


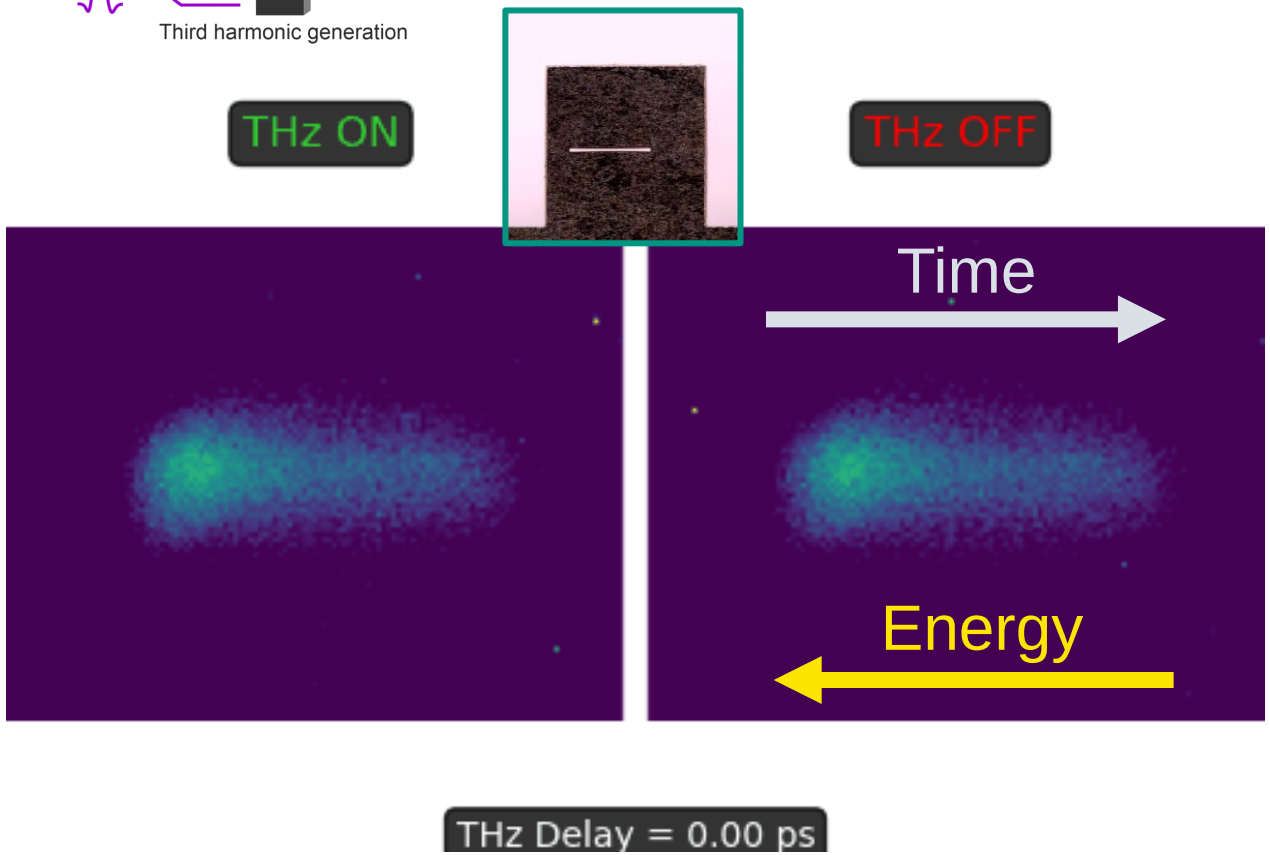
Puzzle complete: Streaking!



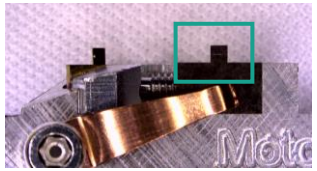


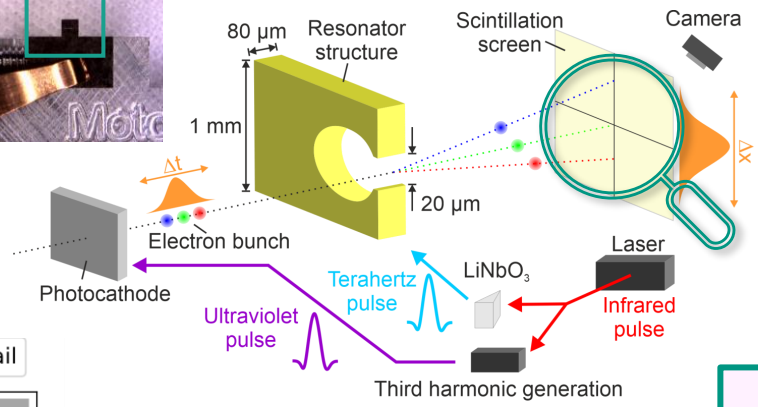


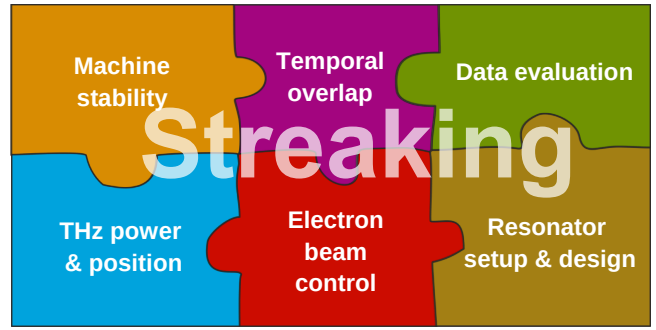


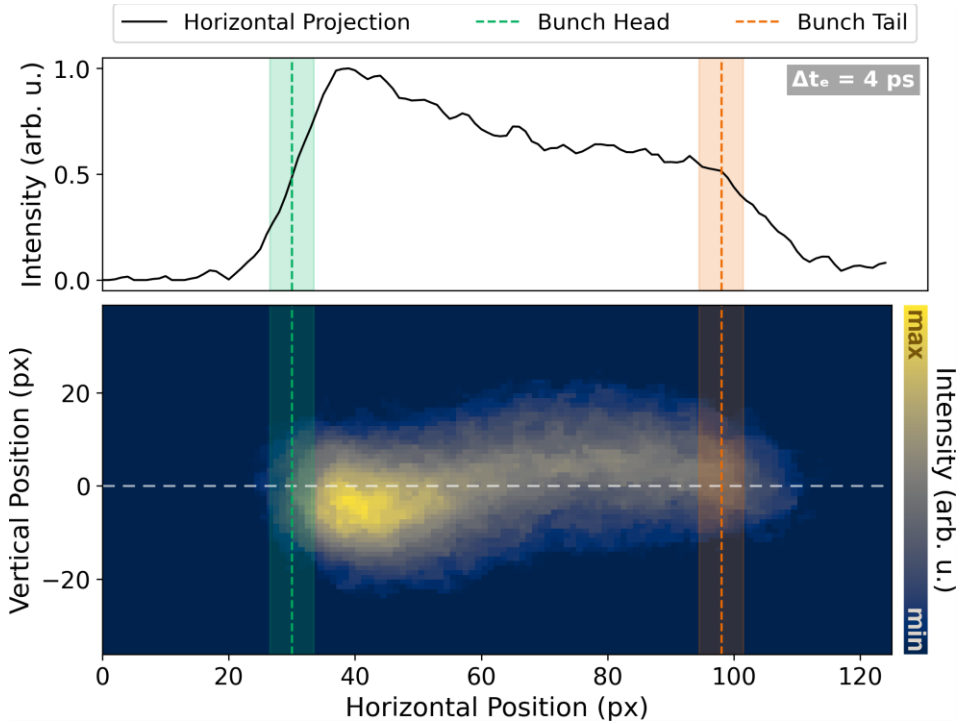


Puzzle complete: Streaking!

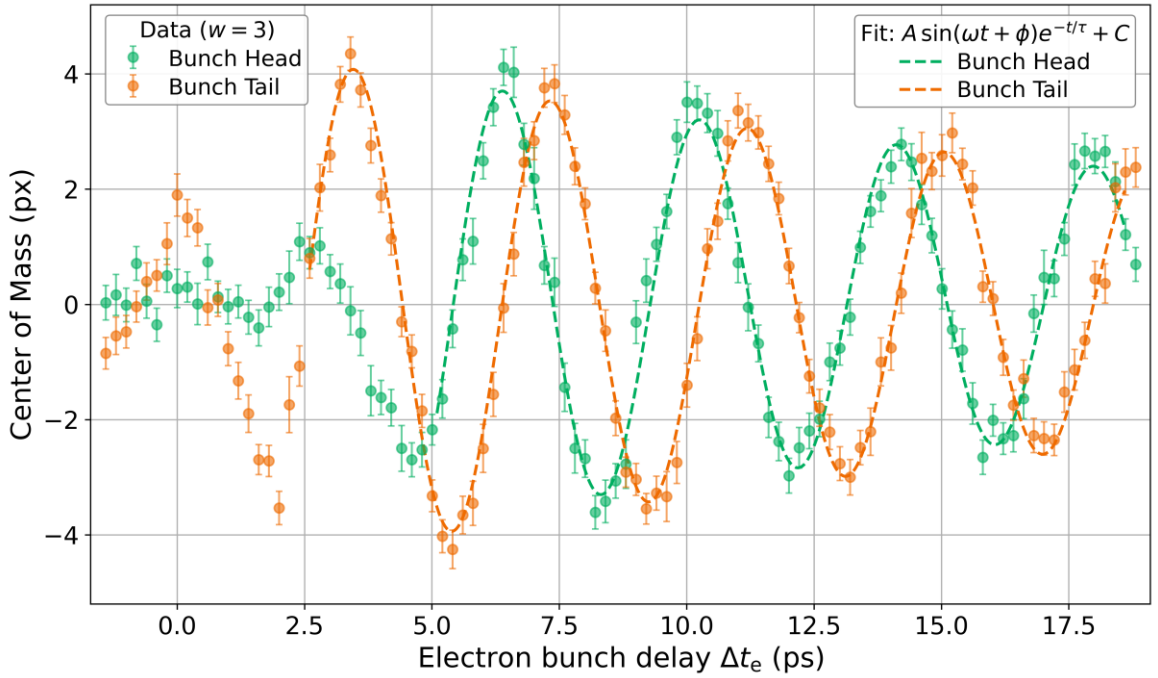












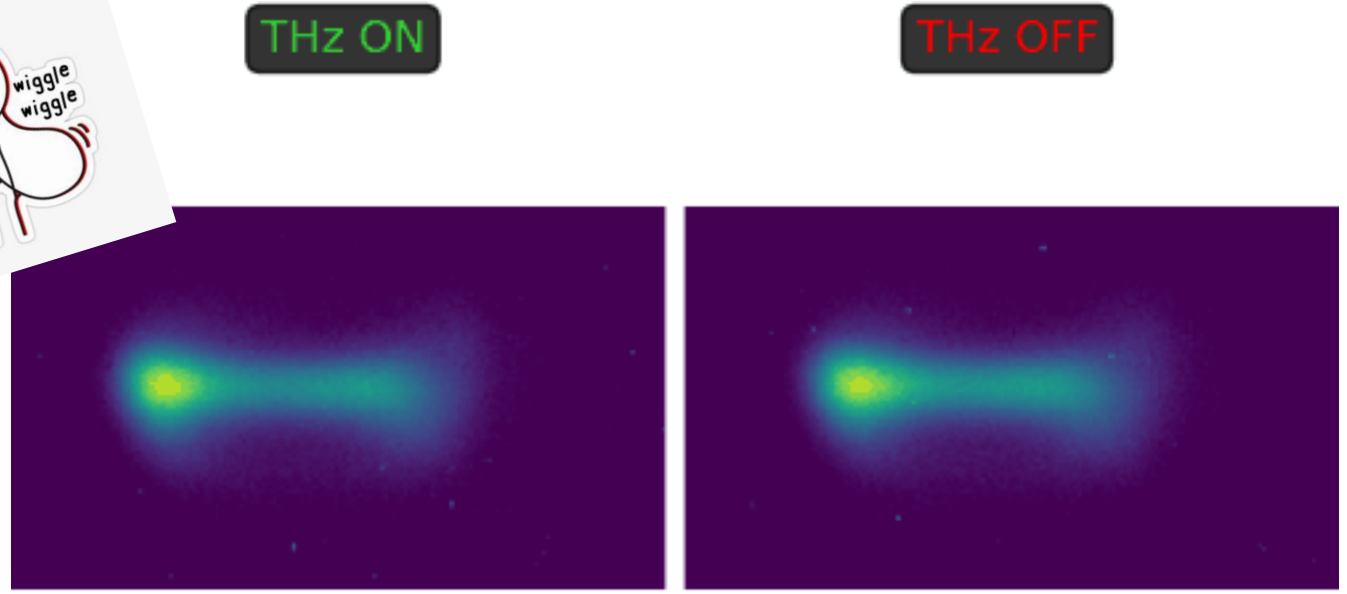
Preliminary + assumptions:

FLUTE e⁻ bunch length → τ_{bunch} = 2.875 ± 0.019 ps (FWHM)

Matthias Nabinger
Doctoral researcher

Contact: matthias.nabinger@kit.edu





Thank you for your attention!

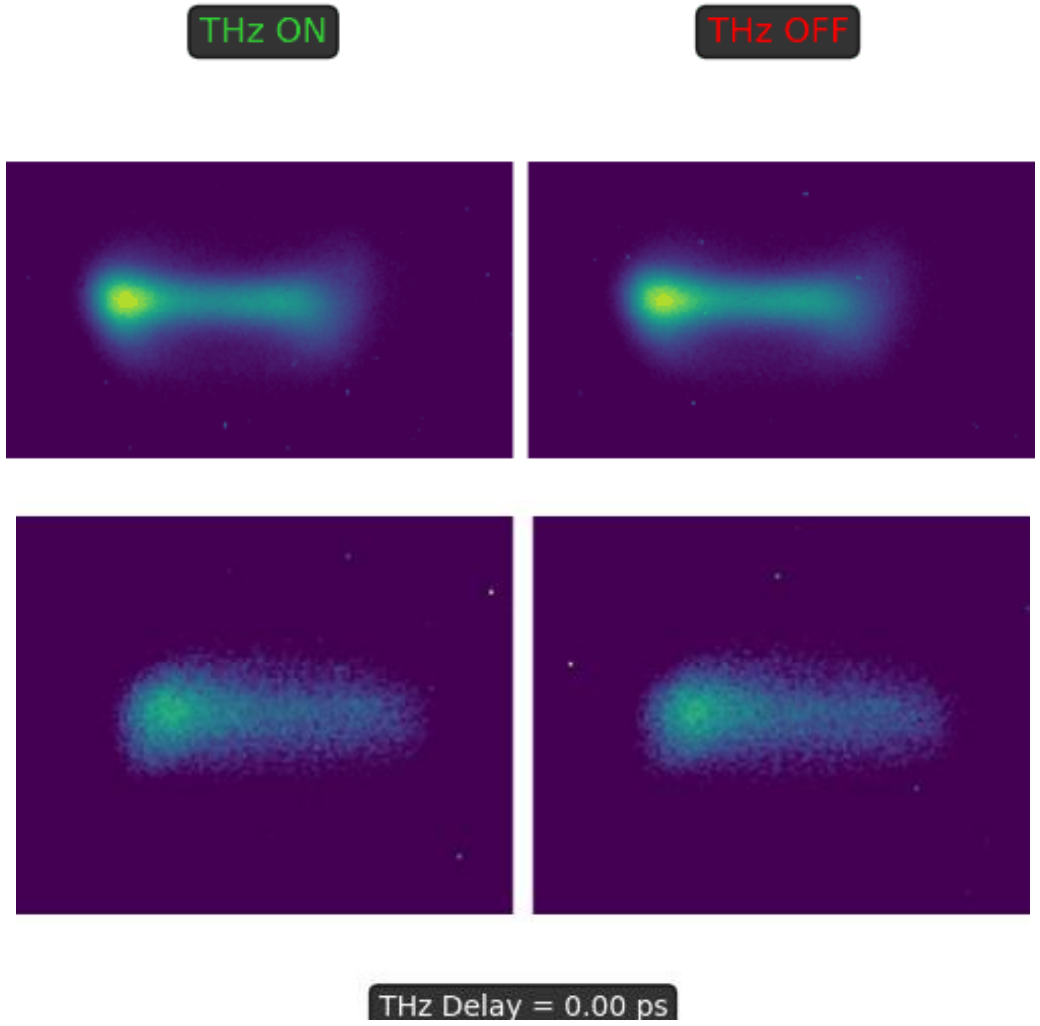
On behalf for the Compact-TDS collaboration:
M. Nabinger, M. J. Nasse, J. Schäfer, E. Bründermann, M. Fuchs,
A. Malygin, M. Noll, J. Steinmann, R. Ruprecht, T. Schmelzer,
M. Schuh, N. J. Smale, A.-S. Müller (KIT)
M. M. Dehler, R. Ischebeck, M. Moser, V. Schlott (PSI)
T. Feurer, M. Hayati, Z. Ollmann (Uni Bern)
S. Glukhov, O. Boine-Frankenheim (TU Darmstadt)

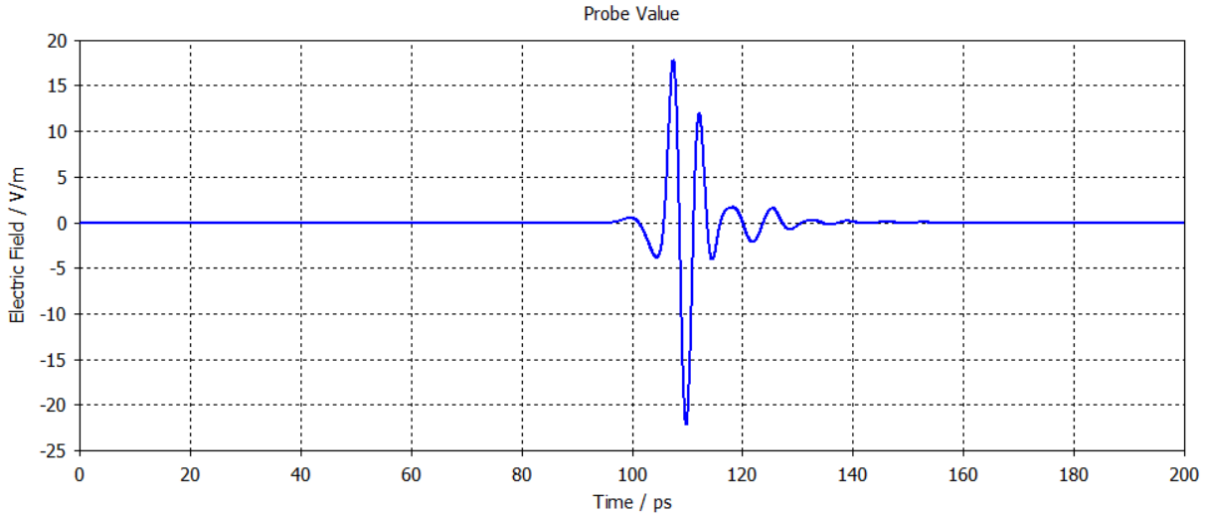
Bundesministerium
für Bildung
und Forschung

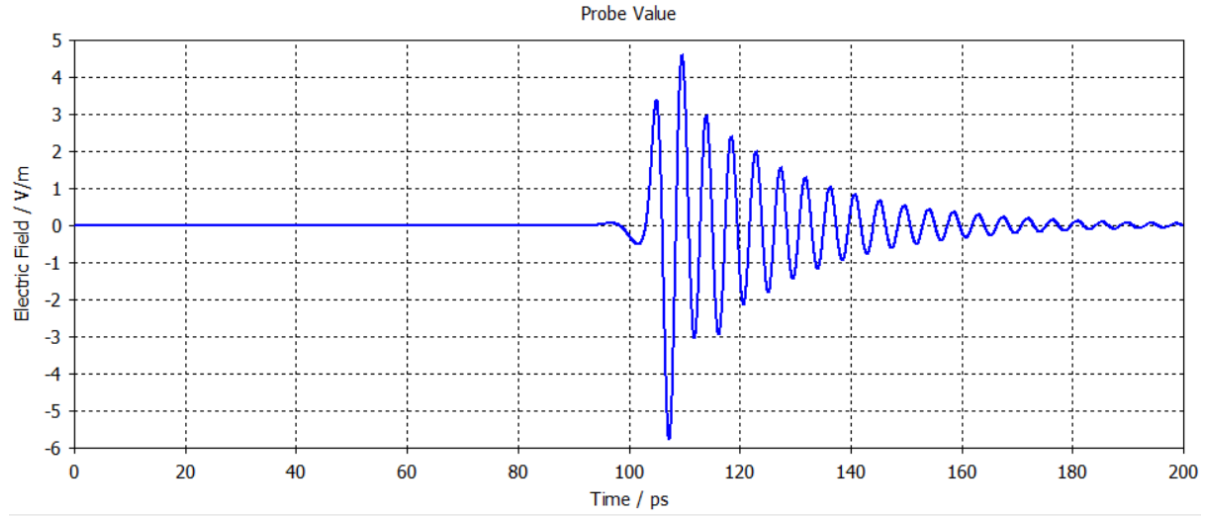
FKZ 05K22VK3

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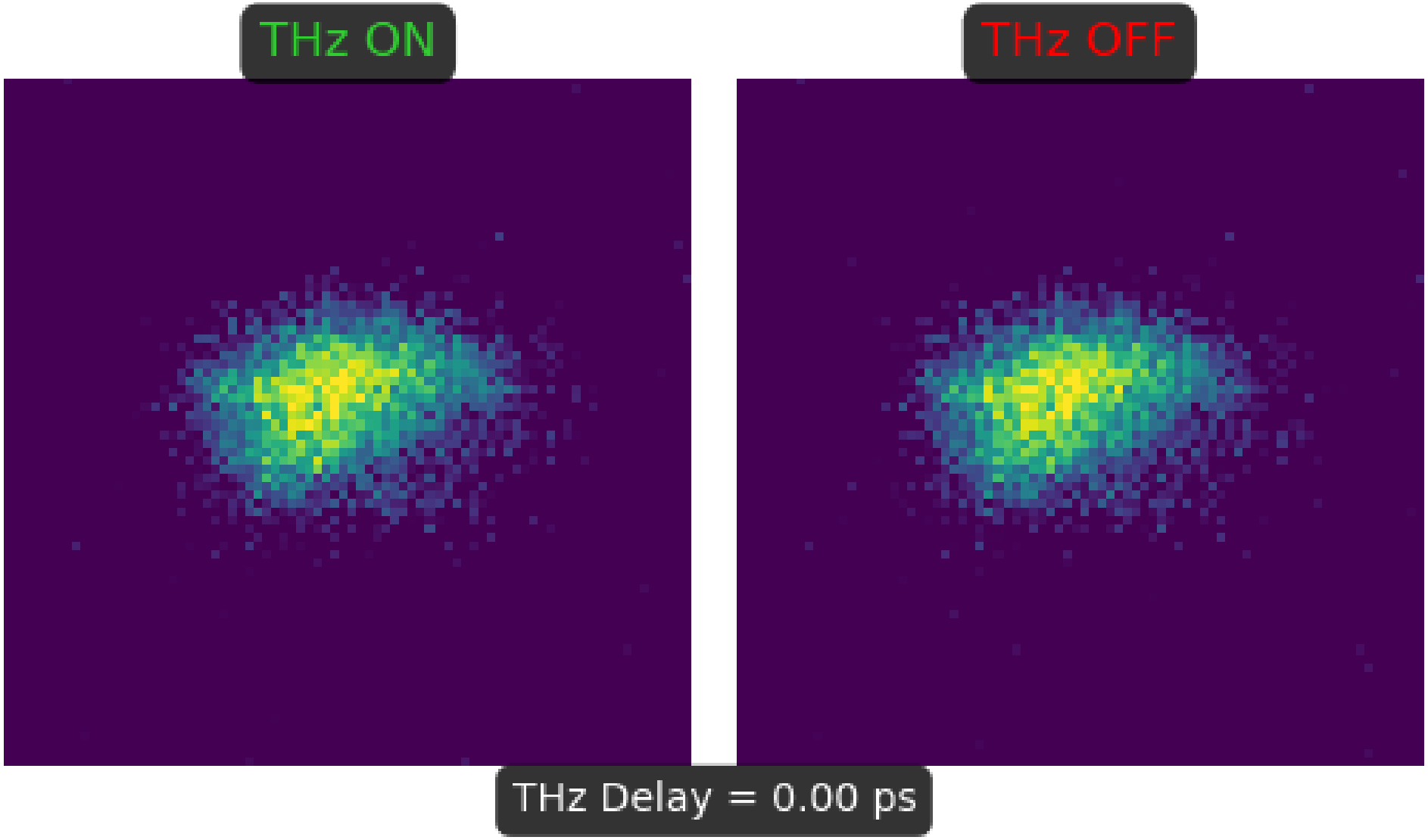
Backup


THz ON
THz OFF
THz Delay = 0.00 ps

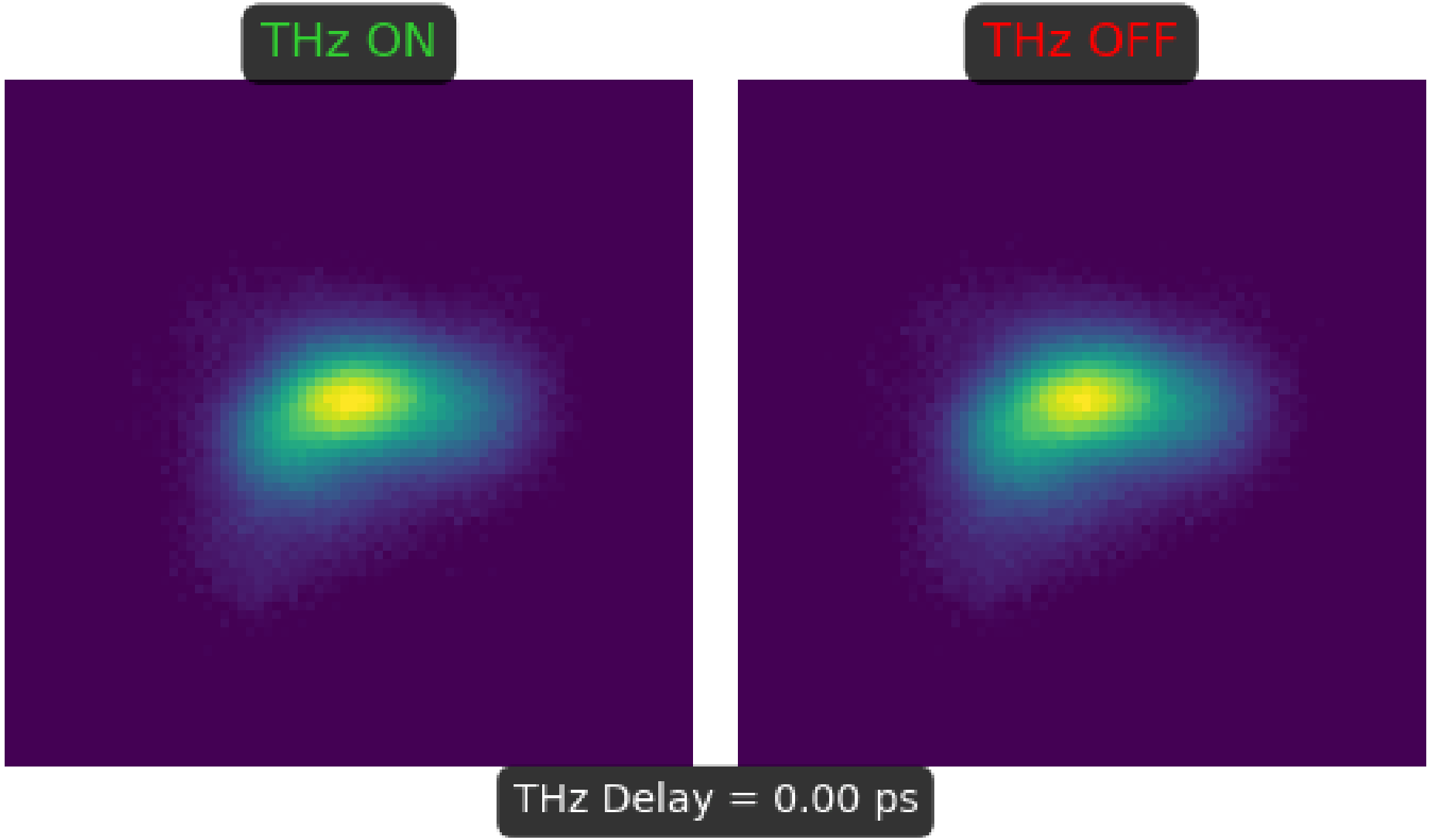

Probe Value
Electric Field / V/m
Time / ps


Probe Value
Electric Field / V/m
Time / ps

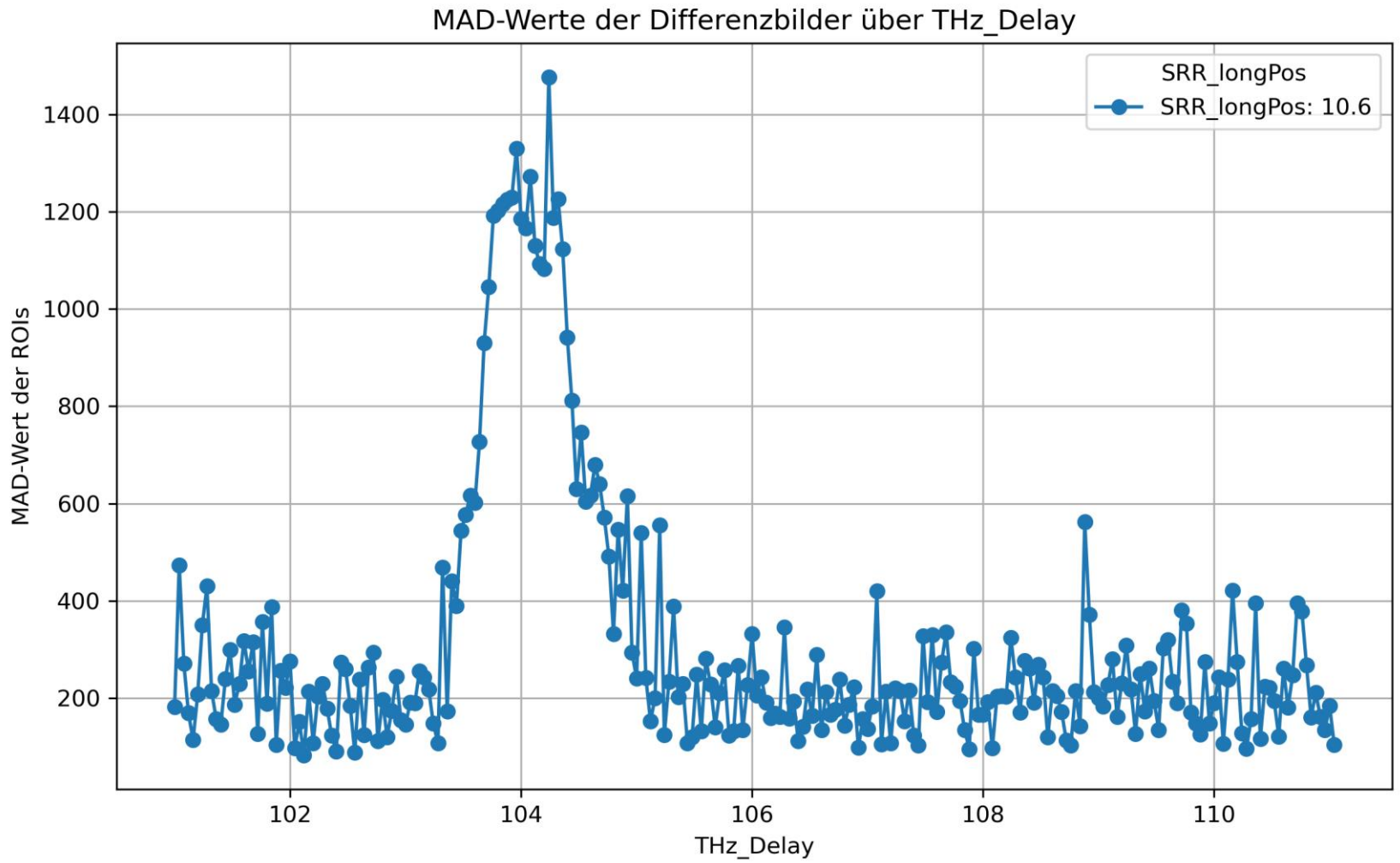
Backup



Backup



Backup


MAD-Werte der Differenzbilder über THz_Delay
SRR_longPos
SRR_longPos: 10.6
MAD-Wert der ROIs
1400
1200
1000
800
600
400
200
102
104
106
108
110
THz_Delay